Algorithmic Fairness in Student Performance Prediction: Detecting and Mitigating Bias Across 32,000 Online Learners

PROJECT REPORT

SUBMITTED BY

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UNDER THE GUIDANCE OF

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In

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SELF BONAFIED CERTIFICATE

This is to certify that the Project Work Report titled ***Algorithmic Fairness in Student Performance Prediction: Detecting and Mitigating Bias Across 32,000 Online Learners*** is a bonafied record of work carried out by **CHALLAPLLI KEERTHANA SAI LASYA PRIYA [EC2532251010088]** under my oversight. It is further certified that, to the best of my knowledge, the work reported herein does not form part of any other internship report or dissertation on the basis of which a degree or award has been conferred on this or any other candidate or any previous occasion.



CHALLAPALLI KEERTHANA SAI LASYA PRIYA

DECLARATION

I hearby declare that the project entitled ***Algorithmic Fairness in Student Performance Prediction: Detecting and Mitigating Bias Across 32,000 Online Learners***

submitted to ***SRM Institute of Science and Technology*** in partial fulfillment of the requirements for the award of the degree of ***Master of Computer Applications (MCA***) is a record of original work carried out by me.

I further declare that this work has not been submitted previously for the award of any degree, diploma, or other qualification in this or any other institution.

I also confirm that the project has been developed independently by me without any external assistance, except for the references mentioned in this report.



CHALLAPALLI KEERTHNA SAI LASYA PRIYA

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I would like to express my sincere gratitude to **SRM Institute of Science and Technology** for providing me with the opportunity to undertake this project.

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CHALLAPALLI KEERTHANA SAI LASYA PRIYA

ABSTRACT

Machine learning-based student outcome prediction has become a cornerstone of modern educational technology, with institutions worldwide deploying automated systems to identify at-risk students and allocate academic support. However, the fairness of these systems - whether they treat students from all demographic groups equitably - has received comparatively little scrutiny. This project addresses this gap by developing, evaluating, and critically auditing a fairness-aware machine learning pipeline for student outcome prediction using the Open University Learning Analytics Dataset (OULAD).

The OULAD dataset comprises 32,593 student enrolments across seven course modules at the UK Open University. Two classification models were trained to predict binary pass/fail outcomes: Logistic Regression as an interpretable baseline achieving 87.2% accuracy and 0.942 ROC-AUC, and XGBoost as the primary model achieving 90.2% accuracy and 0.966 ROC-AUC.

The project makes three primary research contributions. First, it identifies and quantifies intersectional disadvantage: students who are simultaneously disabled and from the lowest socioeconomic backgrounds (IMD band 0-20%) are predicted to pass at just 32.6% compared to 51.3% for all other students - an 18.7 percentage point gap entirely invisible when demographic groups are examined in isolation. The disability gap alone is 10.2 pp;

the IMD gap alone is 17.8 pp. The intersectional gap exceeds both, confirming non-additive compounding of disadvantage invisible to standard single-attribute fairness analysis.

Second, the project rigorously evaluates Fairlearn-based bias mitigation techniques. Threshold Optimizer reduced equalized odds difference for disability from 0.044 to 0.007 - an 84.1% reduction - at less than 0.3% accuracy cost. A methodological correction is also introduced: prior OULAD research applied disability-trained mitigators to IMD fairness evaluation, producing misleading results. This project demonstrates that IMD fairness must be evaluated with IMD-trained mitigators, which reveal a persistent demographic parity gap of 0.162.

Third, a SHAP ensemble-averaged proxy-shift audit reveals a counterintuitive finding: post-mitigation SHAP values increased for every feature including protected attributes. Disability SHAP increased from 0.017 to 1.429 - a 8,289% increase. The Exponentiated Gradient ensemble achieved balanced error rates by amplifying rather than reducing reliance on demographic information, fundamentally challenging the assumption that fairness metric improvements correspond to genuine model debiasing.

An additional analysis disaggregates withdrawal from academic failure by demographic group. The intersectional subgroup withdraws at 43.65% vs 30.83% for all others, while failure rates differ by only 4 pp - showing that non-academic withdrawal, not underperformance, is the primary driver of outcome disparities.

The system is implemented as a 26-cell reproducible Jupyter notebook in Google Colaboratory using Python 3.10 with pandas, scikit-learn, XGBoost, Fairlearn, and SHAP. All results are fully reproducible with random_state=42. Complete source code is available at**: https://github.com/Lasya-priya-ux/OULAD-Algorithmic-Fairness**.

*(**Keywords**: Educational Data Mining, Algorithmic Fairness, OULAD, XGBoost, Bias Mitigation, Fairlearn, SHAP Explainability, Intersectional Fairness, Student Outcome Prediction.)*

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CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

The rapid expansion of online and distance education has produced unprecedented quantities of data about student learning behaviour. Institutions now record every interaction a student makes with their virtual learning environment creating rich longitudinal datasets. Machine learning researchers have used this data to build predictive models that forecast academic outcomes weeks before final results, enabling institutions to target support precisely.

Yet this rapid deployment has outpaced critical evaluation of fairness properties. A growing body of research in algorithmic fairness has documented systematic bias in predictive systems across high-stakes domains. Educational AI has begun to attract similar scrutiny: prediction models trained on historical educational data consistently disadvantage students from protected groups including those with disabilities and those from low socioeconomic backgrounds.

The mechanism of bias is not intentional discrimination but rather the encoding of historical inequities in training data. Students with disabilities have historically received inadequate accommodations; students from low-income backgrounds have faced greater financial barriers to engagement. These inequities appear in training data as statistical associations between demographic characteristics and poor outcomes. A model trained on this data will reproduce them in its predictions.

This project builds on this growing literature by developing a complete, end-to-end fairness-aware machine learning pipeline on the Open University Learning Analytics Dataset and conducting the most rigorous fairness audit to date on this dataset, including the first intersectional analysis of the disability x low-IMD subgroup.

1.2 Problem Statement

Despite significant research in educational AI and algorithmic fairness, several important questions remained unanswered before this project. While single-attribute bias on OULAD had been documented, the intersectional case - students who are both disabled and from low-income backgrounds - had not been examined. Additionally, prior work made a methodological error: applying disability-trained mitigators to evaluate IMD band fairness. Most fundamentally, no prior OULAD study verified whether fairness metric improvements correspond to genuine changes in how the model uses protected demographic information.

1.3 Aim and Objectives

The primary aim is to develop a fairness-aware machine learning pipeline for predicting student academic outcomes and to rigorously evaluate both predictive and equity properties.

- Construct a complete preprocessing pipeline merging seven OULAD CSV files into a single analysis-ready dataset of 32,593 rows.
- Train and evaluate Logistic Regression and XGBoost classifiers using accuracy, precision, recall, F1 score, and ROC-AUC.

- Quantify single-attribute algorithmic bias for disability status and IMD band using demographic parity difference and equalized odds difference.
- Quantify intersectional bias for the disability x low-IMD subgroup and compare it to single-attribute measures.
- Apply Exponentiated Gradient and Threshold Optimizer mitigation using Fairlearn with correctly matched sensitive feature specification.
- Conduct an ensemble-averaged SHAP explainability audit to detect proxy-shift behaviour following bias mitigation.
- Disaggregate withdrawal from academic failure by demographic group and analyse implications for outcome target construction.

1.4 Scope

This project analyses the OULAD dataset from the UK Open University covering 32,593 enrolments across seven course modules. Two protected attributes are examined: disability status (binary) and Index of Multiple Deprivation band (ordinal, ten deciles). The intersectional analysis covers the disability x low-IMD (0-20%) intersection comprising 843 students. The system is implemented as a research pipeline and does not constitute a production deployment recommendation.

1.5 Report Organisation

Chapter 2 reviews relevant literature on algorithmic fairness, educational AI bias, intersectionality in machine learning, and SHAP explainability. Chapter 3 provides problem

analysis, dataset description, UML diagrams, and system requirements. Chapter 4 covers system design. Chapter 5 details the implementation of all six pipeline modules with full code listings. Chapter 6 presents all results and visualisations. Chapter 7 documents the testing approach and results. Chapter 8 describes tools and technologies. Chapter 9 concludes with findings, limitations, and future work.

CHAPTER 2

LITERATURE REVIEW

2.1 Foundations of Algorithmic Fairness

The formal study of algorithmic fairness emerged following high-profile findings that deployed AI systems exhibited systematic bias against protected demographic groups. **Dwork et al. (2012)** established the principle in "Fairness through Awareness" that a fair classifier should assign similar outputs to similar individuals regardless of group membership. Hardt, Price, and **Srebro (2016)** proposed equalized odds as a tractable group fairness standard, requiring equal true positive rates and false positive rates across protected groups - the primary fairness metric in this project because both error types carry significant educational consequences.

Subsequent theoretical work established fundamental limits. **Chouldechova (2017)** proved that no classification system can simultaneously satisfy calibration within groups, balance for the positive class, and balance for the negative class when base rates differ across groups. This impossibility result implies that choosing a fairness criterion is a normative decision - optimising for one criterion necessarily compromises others. This trade-off is directly relevant to educational AI where different fairness criteria reflect different conceptions of educational equity.

The distinction between individual and group fairness has generated substantial theoretical debate. Critics argue group fairness criteria can require treating individuals differently based

on group membership. Defenders argue that in the presence of historical discrimination, group-level adjustments are necessary to correct for systemic biases embedded in training data - precisely the situation in educational datasets like OULAD.

2.2 Algorithmic Bias in Educational Systems

The application of predictive analytics to educational outcomes traces from early data mining approaches **(Romero and Ventura, 2010)** through the development of learning analytics. The availability of large-scale educational datasets, particularly OULAD **(Kuzilek et al., 2017),** has accelerated research considerably. XGBoost consistently outperforms alternatives including SVMs, Neural Networks, and Naive Bayes on tabular educational data.

Boateng and Boateng (2025) documented systematic bias against students from protected groups in prediction systems, highlighting that models trained on historical institutional data consistently reproduce historical inequities. **Opoku, Pei and Xing (2025)** found a consistent inverse relationship on OULAD between model complexity and fairness: increasing accuracy tended to increase rather than decrease fairness violations - a finding consistent with results in this project.

Gunasekara and Saarela (2025) applied post-processing bias mitigation on OULAD and established baseline fairness benchmarks, although their IMD evaluation methodology contains a systematic error this project corrects: they applied disability-trained mitigators to IMD evaluation, producing misleadingly optimistic IMD improvement results**. El**

Jihaoui et al. (2026) propose an explainability-integrated fairness evaluation framework, motivating the SHAP audit in Module 6 of this project.

2.3 Intersectional Fairness in Machine Learning

Intersectionality was introduced by **Kimberly Crenshaw (1989**) to describe how individuals belonging to multiple marginalised groups experience forms of discrimination that cannot be reduced to the sum of single-group discriminations. **Wang et al. (2022**) demonstrated that standard group fairness metrics examining one attribute at a time are systematically blind to intersectional effects - a problem known as fairness gerrymandering where selection of evaluation subgroups determines apparent fairness.

Foulds et al. (2020) proposed differential fairness metrics specifically designed to detect non-additive intersectional effects. **Gohar and Cheng (2023)** surveyed intersectional fairness in machine learning, identifying central challenges: small subgroup sample sizes, combinatorial explosion of possible intersections, and lack of consensus on which intersections constitute legally protected categories.

Kanubala and Valera (2025) found significant misalignment between legal definitions of intersectional discrimination and statistical metrics: courts define discrimination in terms of counterfactual causation while statistical metrics measure distributional properties not directly corresponding to causal claims. The present project finding that disabled, low-IMD students face an 18.7 pp predicted pass rate gap - invisible in single-attribute analysis - directly demonstrates this gap on real educational data.

2.4 Bias Mitigation Techniques

Bias mitigation techniques are classified by their position in the modelling pipeline. The Exponentiated Gradient algorithm **(Agarwal et al., 2018)** is the current state-of-the-art in-processing method, reframing fairness-constrained classification as a two-player game producing an ensemble of base classifiers with different cost weights. The Threshold Optimizer **(Hardt et al., 2016)** is the leading post-processing technique, deriving group-specific decision thresholds by solving a linear programme optimising accuracy subject to equalized odds constraints.

Aoudi and Al-Aqrabi (2025) found that post-processing methods consistently achieve larger fairness improvements at lower accuracy costs than in-processing alternatives in educational contexts - consistent with the present project results where Threshold Optimizer outperformed Exponentiated Gradient on equalized odds improvement for disability (84.1% vs 36.4% reduction).

2.5 SHAP Explainability and the Proxy-Shift Problem

SHAP **(Lundberg and Lee, 2017)** applies Shapley values from cooperative game theory to attribute each prediction to individual features based on marginal contribution across all feature orderings. TreeExplainer computes exact Shapley values for tree ensembles in polynomial time, making it feasible for large datasets and critical for this project where small differences in protected attribute importance are the focus of analysis.

The proxy-shift problem refers to mitigation techniques reducing direct use of protected attributes while increasing reliance on proxy features carrying similar demographic information. The present project finding - that every feature SHAP value increased post-mitigation including protected attributes themselves - represents a previously undocumented failure mode distinct from standard proxy-shift: amplification of protected attribute reliance rather than transfer to proxies. This is only detectable using the ensemble-averaging methodology correctly representing all eight EG base predictors rather than a single-predictor approximation.

2.6 Research Gap

Four gaps addressed by this project: **(1)** no prior OULAD study quantified intersectional disadvantage for the disability x low-IMD subgroup; **(2)** prior OULAD mitigation studies applied disability-trained mitigators to IMD evaluation - a methodological error this project corrects; **(3)** no prior OULAD study conducted a SHAP explainability audit of mitigation effectiveness; **(4)** no prior OULAD study disaggregated withdrawal and academic failure by intersectional demographic group.

CHAPTER 3

ANALYSIS AND REQUIREMENTS

3.1 Problem Analysis

The core problem is that machine learning models trained on educational data may exhibit systematic bias against students from protected demographic groups, producing predictions that disadvantage them relative to the general student population. Three analytical questions drive this project**: (1)** Does bias exist in XGBoost and LR models on OULAD? **(2**) Is intersectional bias worse than single-attribute bias for the disability x low-IMD subgroup? **(3)** Do Fairlearn mitigation techniques genuinely reduce protected attribute reliance or only improve surface metrics?

3.2 Dataset Analysis

The Open University Learning Analytics Dataset (OULAD) released by **Kuzilek et al. (2017)** comprises seven interrelated CSV files covering 32,593 student enrolments across seven course modules over multiple presentation periods in 2013 and 2014.

File	Rows	Columns	Description
studentInfo.csv	32,593	12	Student demographics and final results
studentAssessment.csv	173,912	5	Individual assessment submission scores
studentVle.csv	10,655,280	6	Daily VLE clickstream interactions
assessments.csv	206	6	Assessment metadata
studentRegistration.csv	32,593	5	Registration and withdrawal dates
vle.csv	6,364	6	VLE activity type metadata
courses.csv	22	3	Module and presentation metadata

Table 3.1: OULAD Dataset Files

The studentInfo file contains twelve variables per student including gender (17,875 male / 14,718 female), disability status (3,164 disabled / 29,429 non-disabled), IMD band (ten deciles), age band, highest education, number of previous attempts, studied credits, and final_result (Pass 37.9%, Withdrawn 31.2%, Fail 21.6%, Distinction 9.3%). The studentVle file with 10,655,280 rows records every click a student made in the virtual learning environment, constituting the source of the total_clicks behavioural predictor.

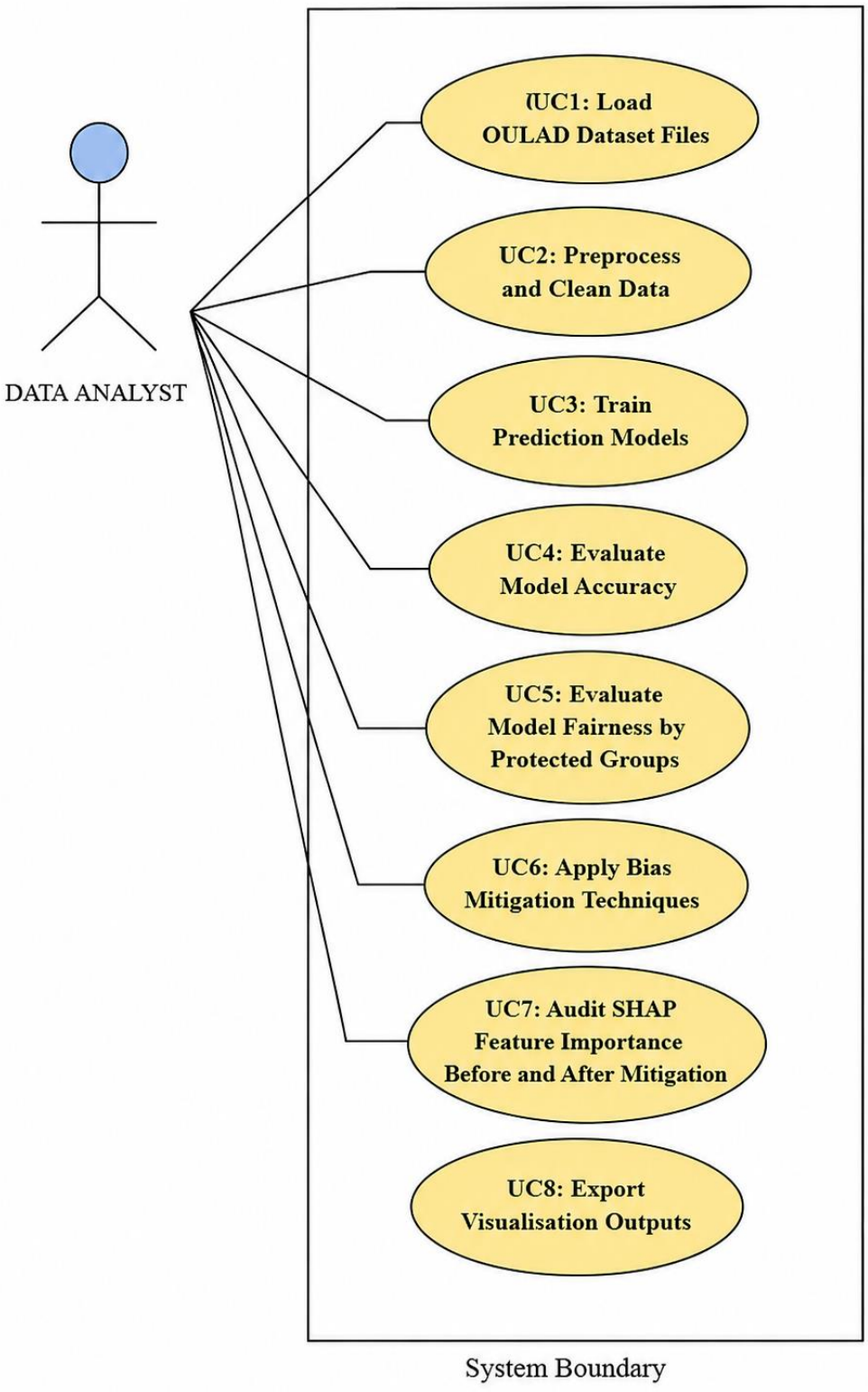
3.3 UML Analysis Models

3.3.1 Use Case Diagram

Primary Actor: Data Analyst. External Entity: OULAD Dataset.

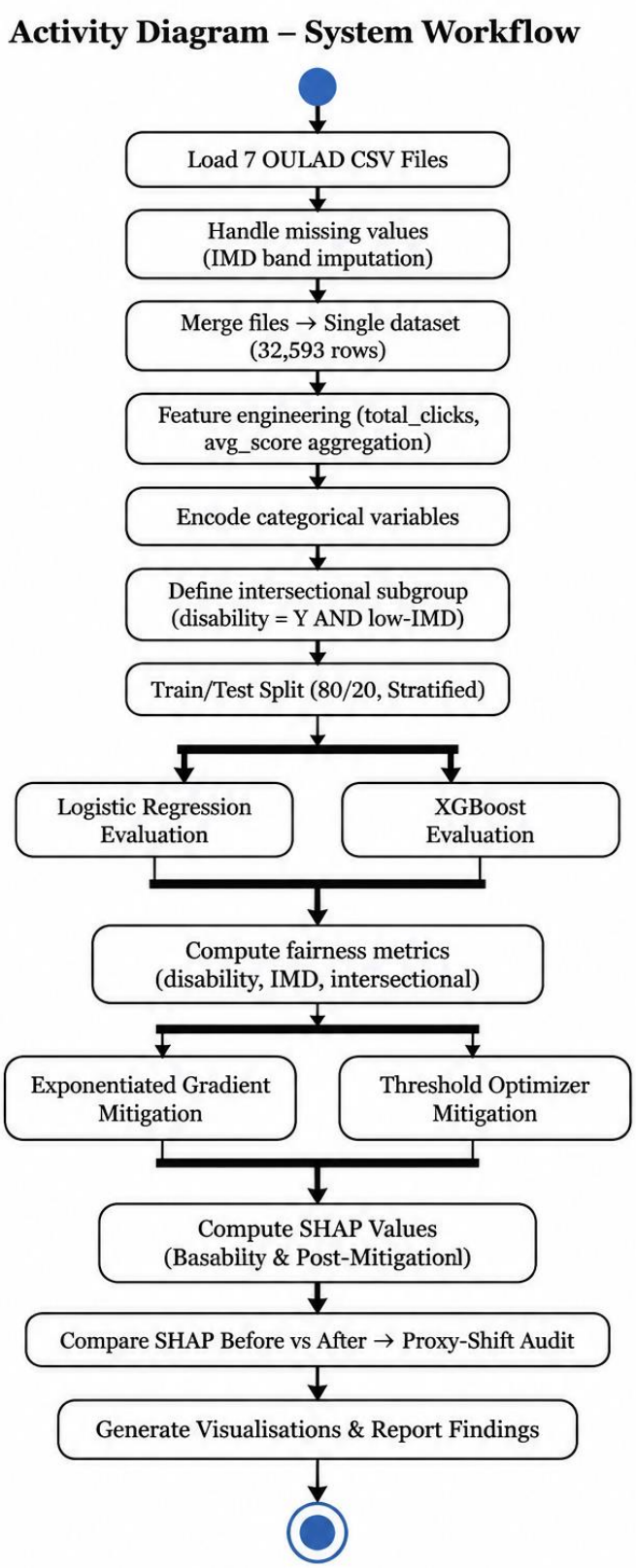
- **UC1: Load OULAD Dataset Files** - Load and validate all seven CSV files, check shapes against expected dimensions.
- **UC2: Preprocess and Clean Data** - Handle missing values, encode features, construct binary target, define intersectional subgroup.
- **UC3: Train Prediction Models** - Train LR with StandardScaler and XGBoost on 80% stratified split.
- **UC4: Evaluate Model Accuracy** - Compute accuracy, precision, recall, F1 score, and ROC-AUC on held-out test set.
- **UC5: Evaluate Model Fairness** - Compute DPD and EOD for disability, IMD band, and intersectional subgroup.

- **UC6: Apply Bias Mitigation** - Apply EG and TO using Fairlearn with correctly matched sensitive features.
- **UC7: Audit SHAP Feature Importance** - Compute ensemble-averaged SHAP values; detect proxy-shift behaviour.
- **UC8: Export Visualisation Outputs** - Save four PNG charts at 150 DPI minimum.


DATA ANALYST
(UC1: Load OULAD Dataset Files
UC2: Preprocess and Clean Data
UC3: Train Prediction Models
UC4: Evaluate Model Accuracy
UC5: Evaluate Model Fairness by Protected Groups
UC6: Apply Bias Mitigation Techniques
UC7: Audit SHAP Feature Importance Before and After Mitigation
UC8: Export Visualisation Outputs
System Boundary

3.3.2 Activity Diagram

- Load 7 CSV files and validate shapes.
- Impute 1,111 missing IMD values using modal category (mode = 20-30%).
- Aggregate VLE: 10,655,280 rows into 29,228 per-student totals using pandas groupby.
- Aggregate assessments: 173,912 rows into 25,843 per-student averages.
- Three-table merge producing 32,593 rows x 16 columns.
- Encode five categorical variables; construct binary pass_fail target.
- Define intersectional_group variable (843 students satisfying disability=Y AND imd_band in 0-10% or 10-20%).
- Save oulad_clean.csv (3.6 MB, 32,593 x 22 columns).
- 80/20 stratified split; train LR and XGBoost; evaluate on test set.
- Compute DPD and EOD for three protected attribute configurations.
- Apply EG and TO with disability as sensitive feature; repeat with IMD band.
- Compute baseline SHAP and ensemble-averaged EG SHAP; compare; detect proxy-shift.
- Generate four visualisations and compile report findings.


Activity Diagram – System Workflow
Load 7 OULAD CSV Files
Handle missing values (IMD band imputation)
Merge files → Single dataset (32,593 rows)
Feature engineering (total_clicks, avg_score aggregation)
Encode categorical variables
Define intersectional subgroup (disability = Y AND low-IMD)
Train/Test Split (80/20, Stratified)
Logistic Regression Evaluation
XGBoost Evaluation
Compute fairness metrics (disability, IMD, intersectional)
Exponentiated Gradient Mitigation
Threshold Optimizer Mitigation
Compute SHAP Values (Basablity & Post-Mitigationl)
Compare SHAP Before vs After → Proxy-Shift Audit
Generate Visualisations & Report Findings

3.3.3 Class Diagram

Class	Attributes	Methods
DataLoader	file_paths, raw_dataframes	load_files(), merge_tables()
Preprocessor	df, missing_strategy	impute_missing(), encode_features(), create_target()
ModelTrainer	X_train, y_train, model_type	train(), evaluate(), predict()
FairnessEvaluator	y_true, y_pred, sensitive_feature	compute_dpd(), compute_eod(), group_metrics()
BiasmitigATOR	model, constraint, sensitive_feature	fit(), predict(), compare_metrics()
SHAPAuditor	baseline_model, mitigated_model, X_test	compute_shap(), compare_importance(), detect_proxy_shift()

Table 3.2: Class Diagram

3.4 System Requirements

3.4.1 Functional Requirements

- **FR1**: Load and merge all seven OULAD CSV files into a single 32,593-row dataset.
- **FR2**: Handle missing IMD band values using modal imputation validated by sensitivity analysis.
- **FR3:** Construct binary pass_fail target (Pass+Distinction=1; Fail+Withdrawn=0).
- **FR4:** Aggregate 10,655,280 VLE records into per-student click totals using efficient groupby operations.

- **FR5**: Train LR and XGBoost; evaluate with accuracy, precision, recall, F1, and ROC-AUC.
- **FR6**: Compute DPD and EOD for disability, IMD band, and intersectional subgroup.
- **FR7**: Apply EG and TO mitigation using Fairlearn with correctly matched sensitive features.
- **FR8**: Compute ensemble-averaged SHAP values before and after EG mitigation.
- **FR9**: Generate four visualisation outputs as PNG at 150 DPI minimum.

3.4.2 Non-Functional Requirements

- **NFR1:** System runs on Google Colaboratory with no local hardware installation required.
- **NFR2**: Preprocessing pipeline completes within 10 minutes on standard Colab CPU.
- **NFR3**: All cells documented with inline comments explaining each step.
- **NFR4**: Full reproducibility - random_state=42 throughout ensures identical results across runs.
- **NFR5**: Visualisations saved as PNG at minimum 150 DPI.
- **NFR6**: Intermediate dataset saved as CSV enabling independent re-execution of downstream cells.

3.4.3 Software Requirements

TOOL/LIBRARY	VERSION	PURPOSE
Python	3.10	Primary Programming Language
Google Colaboratory	—	Development and execution
pandas	2.x	Data loading, merging, manipulation
numpy	1.x	Numerical computation
scikit-learn	1.x	Model training, evaluation, preprocessing
XGBoost	2.x	Primary prediction model
Fairlearn	0.10.x	Bias mitigation and fairness metrics
SHAP	0.45.x	Explainability and proxy-shift audit
matplotlib	3.x	Visualisation

Table 3.3: Software Requirements

3.4.4 Hardware Requirements

- **Minimum:** Any device with a modern web browser capable of running Google Colaboratory.
- **Recommended**: Stable internet connection for Colab CPU access.
- **Dataset storage**: Minimum 500 MB free space for OULAD archive and intermediate files.
- **Memory:** The studentVle aggregation requires approximately 4 GB RAM - standard on Google Colab.

CHAPTER 4

SYSTEM DESIGN

4.1 Architectural Design

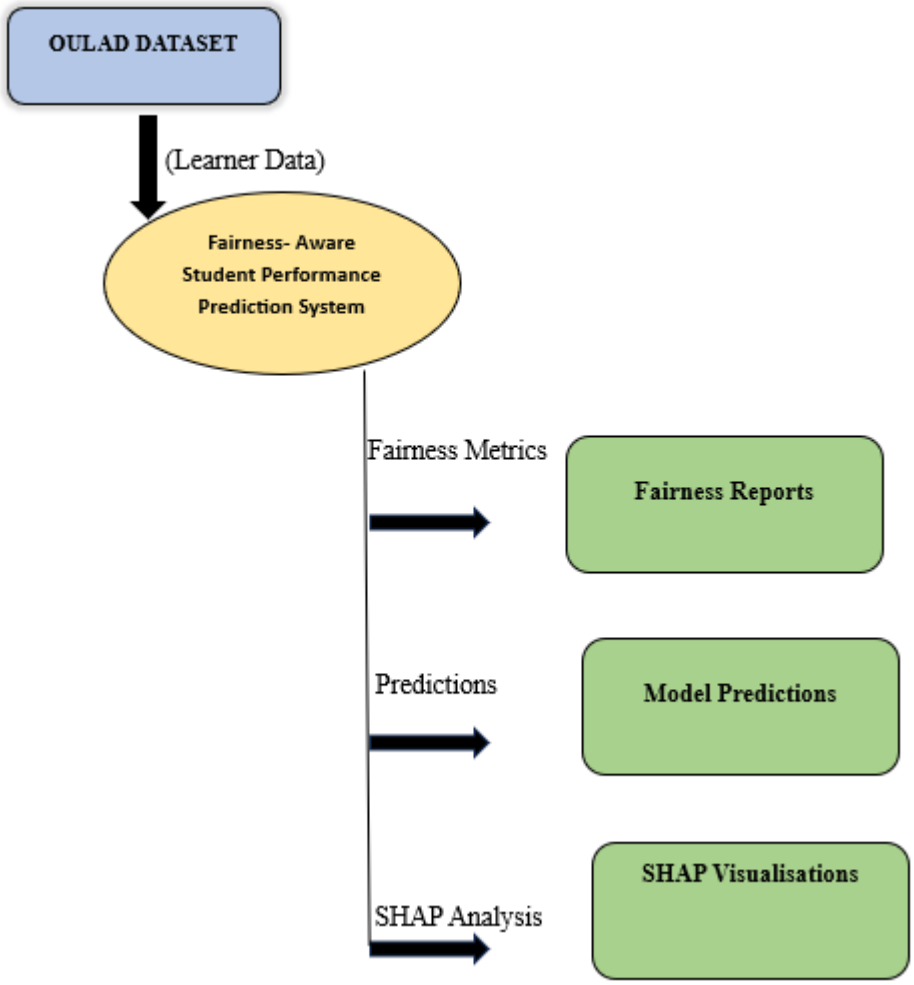
The system follows a linear six-stage pipeline architecture where each stage receives processed data from the previous stage and passes its output forward. Implemented as a reproducible Jupyter notebook in Google Colaboratory with no user-facing web interface. Outputs are stored as flat CSV files and PNG exports. Three design principles govern the architecture: reproducibility (random_state=42 throughout), modularity (each stage independently re-runnable via saved oulad_clean.csv), and transparency (all intermediate outputs printed inline).

- **Stage 1** - Data Ingestion: Loads seven OULAD CSV files and performs three relational joins producing 32,593 rows x 16 columns.
- **Stage 2** - Preprocessing: Handles 1,111 missing IMD values, constructs binary target, engineers aggregate features, encodes five categorical variables, defines intersectional subgroup (843 students). Saves oulad_clean.csv.
- **Stage 3** - Model Training: Trains LR with StandardScaler and XGBoost on 80/20 stratified split. Evaluates on held-out test set of 6,519 records.
- **Stage 4** - Fairness Evaluation: Computes DPD and EOD for disability, IMD band, and intersectional subgroup.

- **Stage 5** - Bias Mitigation: Applies EG and TO with both disability and IMD band as sensitive features.
- **Stage 6** - SHAP Audit: Computes ensemble-averaged SHAP values before and after mitigation; generates proxy-shift visualisation.

4.2 Level 0 DFD (Context Diagram)

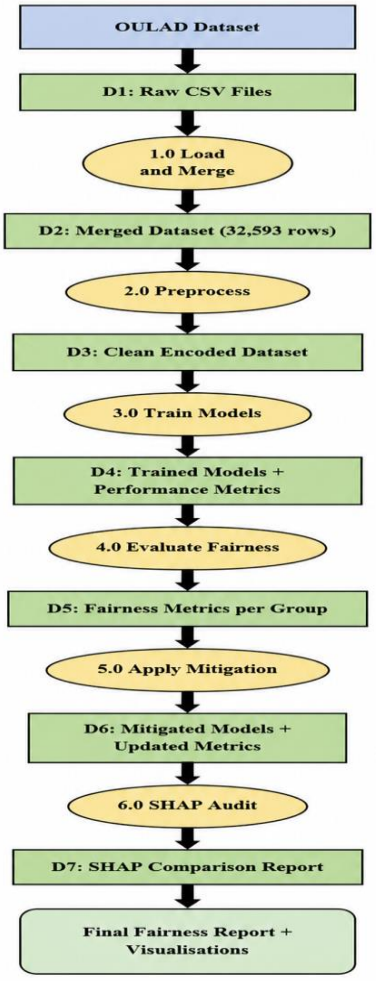
One external entity (OULAD Dataset) provides all input. Three output categories**: (1)** Model Predictions - binary pass/fail for each of 6,519 test set students; **(2)** Fairness Reports - DPD and EOD for all three protected attribute configurations, baseline and post-mitigation; **(3)** SHAP Visualisations - feature importance comparison and proxy-shift analysis.



4.3 Level 1 DFD

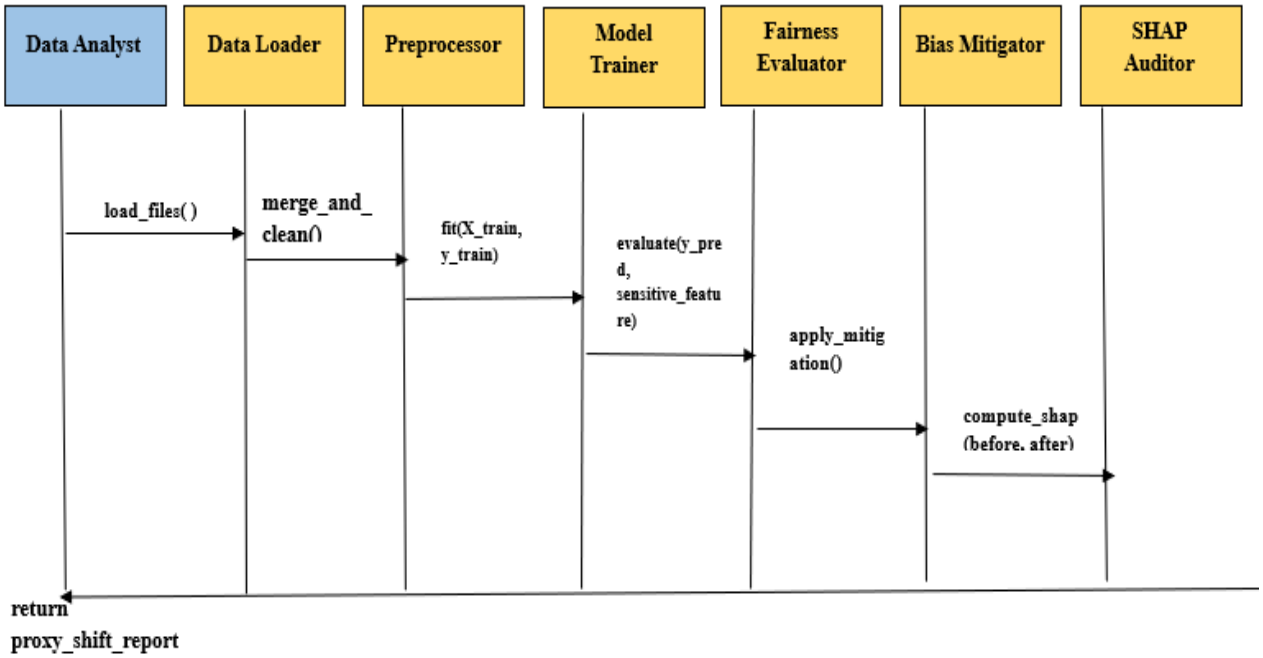
- OULAD Dataset => [D1: Raw CSV Files] => (1.0 Load and Merge) => [D2: Merged Dataset 32,593 rows x 16 cols]
- [D2] => (2.0 Preprocess) => [D3: oulad_clean.csv - 32,593 rows x 22 cols]
- [D3] => (3.0 Train Models) => [D4: Trained LR and XGBoost + 5-metric performance report]
- [D4] => (4.0 Evaluate Fairness) => [D5: DPD and EOD per group - 3 configurations]

- [D5] => (5.0 Apply Mitigation) => [D6: Mitigated Models (EGx2, TOx2) + Updated Metrics]
- [D6] => (6.0 SHAP Audit) => [D7: Ensemble-averaged SHAP Comparison + Proxy-Shift Report]
- [D7] => Output: Final Report + 4 PNG Visualisation Exports



4.4 Sequence Diagram

- **1. DataAnalyst** => DataLoader: load_files() - seven CSVs loaded, shapes validated
- **2. DataLoader** => Preprocessor: merge_and_clean() - joins, imputation, feature engineering, encoding
- **3. Preprocessor** => ModelTrainer: fit(X_train, y_train) - StandardScaler + LR + XGBoost trained
- **4. ModelTrainer** => FairnessEvaluator: evaluate(y_pred, sensitive_feature) - DPD and EOD computed
- **5. FairnessEvaluator** => BiasMitigator: apply_mitigation() - EG and TO with disability then IMD
- **6. BiasMitigator** => SHAPAuditor: compute_shap(before, after) - TreeExplainer on baseline; ensemble-averaged SHAP on EG
- **7. SHAPAuditor** => DataAnalyst: return proxy_shift_report



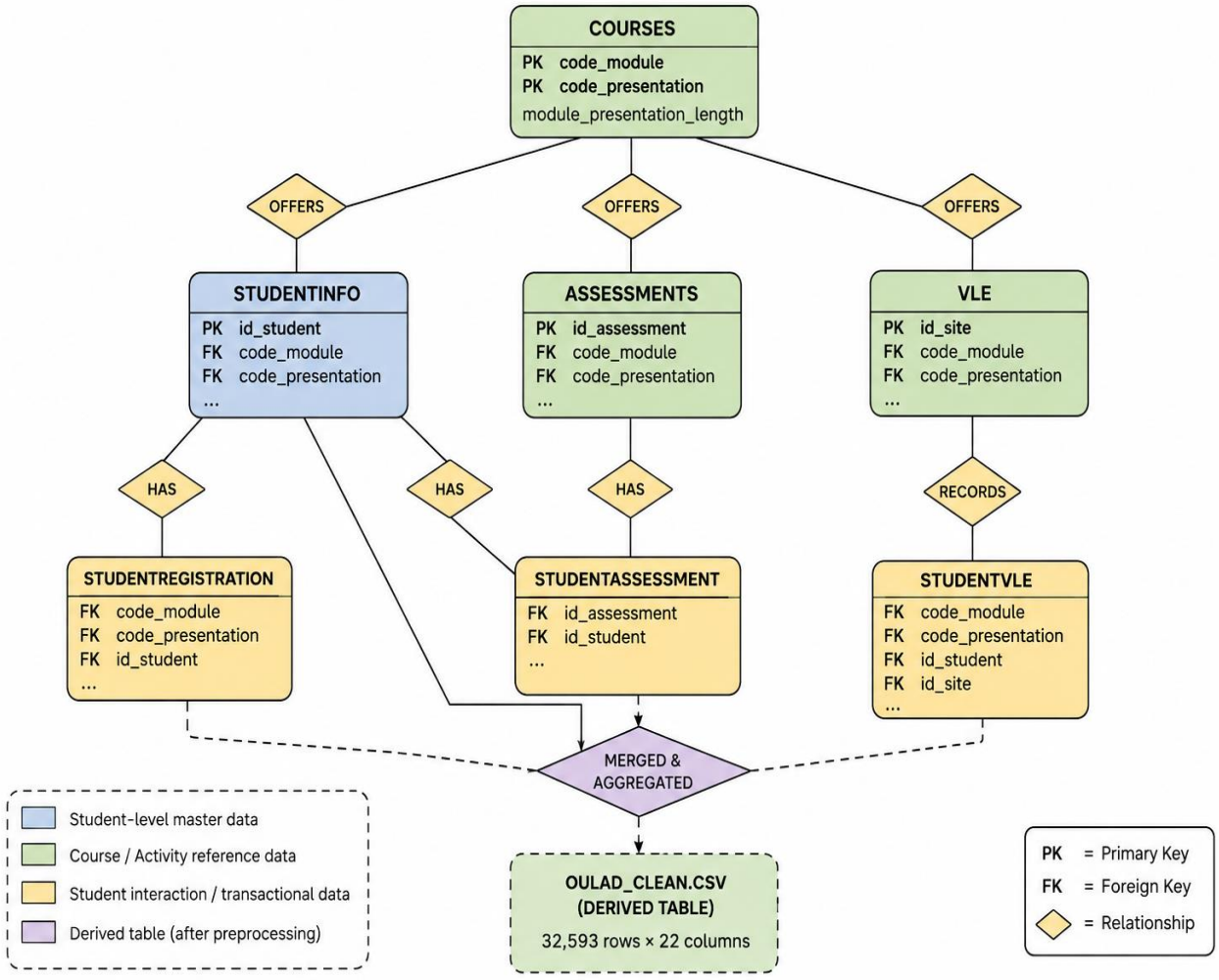
4.5 Database Design

No RDBMS is used. The OULAD dataset is stored and processed as flat CSV files with an explicit relational structure respected during merging. Primary join key: composite (id_student, code_module, code_presentation).

- **studentInfo:** id_student (PK), code_module (FK), code_presentation (FK), gender, region, highest_education, imd_band, age_band, num_of_prev_attempts, studied_credits, disability, final_result
- **studentAssessment:** id_assessment (FK), id_student (FK), date_submitted, is_banked, score
- **assessments:** id_assessment (PK), code_module (FK), code_presentation (FK), assessment_type, date, weight

- **studentVle:** code_module (FK), code_presentation (FK), id_student (FK), id_site (FK), date, sum_click
- **vle:** id_site (PK), code_module (FK), code_presentation (FK), activity_type, week_from, week_to
- **studentRegistration:** code_module (FK), code_presentation (FK), id_student (FK), date_registration, date_unregistration
- **courses:** code_module (PK), code_presentation (PK), module_presentation_length

The derived output table oulad_clean.csv (32,593 x 22 cols, 3.6 MB) contains all demographic variables, engineered features (total_clicks, avg_score, num_assessments), encoded variables, binary target (pass_fail), and intersectional indicator (intersectional_group).



4.6 Design Rationale

4.6.1 XGBoost as Primary Model

XGBoost was selected over Random Forest, Neural Networks, and SVMs for three reasons:

(1) state-of-the-art performance on tabular mixed-type classification**; (2)** native SHAP TreeExplainer compatibility for exact Shapley value computation; **(3)** training time under two minutes on Colab CPU enabling iterative experimentation. Logistic Regression was

retained as an interpretable baseline with directly interpretable log-odds coefficients for sanity-checking feature directions.

4.6.2 Fairlearn over AIF360

Fairlearn was selected because its API integrates cleanly with scikit-learn-compatible estimators, its EG implementation supports XGBoost natively, and its Threshold Optimizer requires no model retraining. AIF360 would require TensorFlow integration or non-trivial API adaptation for XGBoost compatibility.

4.6.3 Equalized Odds as Primary Fairness Metric

Demographic parity requires equal positive prediction rates regardless of actual outcome rates. In educational contexts this would mean predicting equal pass rates even when actual rates differ, which could mask genuine support needs. Equalized odds requires equal true positive AND false positive rates - ensuring proportionally equal errors regardless of base rates. This is ethically appropriate for student support allocation because it prevents both over-flagging capable students and under-flagging struggling students.

4.6.4 Pipeline Pattern for Modularity

Each processing stage is self-contained and passes its output to the next stage via the saved oulad_clean.csv intermediate file. Any stage can be re-executed independently after preprocessing, reducing iteration time for model training experiments from the full pipeline runtime of approximately eight minutes to under two minutes.

CHAPTER 5

IMPLEMENTATION

5.1 Implementation Approach

The system is implemented as a single 26-cell Jupyter notebook in Google Colaboratory. All random operations use random_state=42. Intermediate outputs are printed inline. The preprocessed dataset is saved to oulad_clean.csv so downstream stages can run independently without repeating the 10.6M-row VLE aggregation. Three design patterns are used: Pipeline (each stage passes output forward), Strategy (mitigation techniques as interchangeable parameters), and Facade (FairnessEvaluator encapsulates group-stratified metric computation).

5.2 Module 1 - Data Ingestion and Merging

5.2.1 VLE Aggregation: Efficient large-file aggregation:

The studentVle file contains 10,655,280 rows. Rather than iterating row by row, the implementation uses a single pandas groupby operation to aggregate all clicks per student per module in one pass

```
# === CELL 6: Merge all files into one clean dataset ===

# Step 1: Aggregate VLE clickstream - total clicks per student per module
print("Aggregating VLE clicks (10M rows -> one row per student)...")
vle_agg = student_vle.groupby(
    ['id_student', 'code_module', 'code_presentation']
)['sum_click'].sum().reset_index()
vle_agg.columns = ['id_student', 'code_module', 'code_presentation', 'total_clicks']
print(f"VLE aggregated shape: {vle_agg.shape}")
print()
```

5.2.2 Assessment Aggregation

```
Aggregating VLE clicks (10M rows -> one row per student)...
VLE aggregated shape: (29228, 4)

Aggregating assessment scores...
Assessment aggregated shape: (25843, 5)
```

This reduces 10.6 million records to 29,228 rows in under 30 seconds on Colab CPU

5.3 Module 2 - Preprocessing and Feature Engineering

5.3.1 Missing Value Imputation

```
# === CELL 5: Handle missing values + create binary target ===

# Step 1: Fix the 1111 missing IMD band values
# We use the mode (most common value) to fill them
imd_mode = student_info['imd_band'].mode()[0]
print(f"Filling {student_info['imd_band'].isna().sum()} missing IMD values with mode: {imd_mode}")
student_info['imd_band'] = student_info['imd_band'].fillna(imd_mode)
print(f"Missing IMD values after fix: {student_info['imd_band'].isna().sum()}")
print()
```

5.3.2 Binary Target Construction

```
# Step 2: Create binary target variable
# Pass + Distinction = 1 (positive outcome)
# Fail + Withdrawn = 0 (negative outcome)
student_info['pass_fail'] = student_info['final_result'].apply(
    lambda x: 1 if x in ['Pass', 'Distinction'] else 0
)
print("=== Binary target variable created ===")
print(student_info['pass_fail'].value_counts())
print()
print(f"Overall pass rate: {student_info['pass_fail'].mean()*100:.1f}%")
print()
```

5.3.3 Categorical Encoding

```
# Step 2: Disability - binary encoding (this is protected attribute 1)
df['disability_enc'] = df['disability'].map({'Y': 1, 'N': 0})

# Step 3: IMD band - ordinal encoding
# 0-10% = most deprived = 1, 90-100% = least deprived = 10
imd_order = {
    '0-10%': 1, '10-20': 2, '20-30%': 3, '30-40%': 4, '40-50%': 5,
    '50-60%': 6, '60-70%': 7, '70-80%': 8, '80-90%': 9, '90-100%': 10
}
df['imd_band_enc'] = df['imd_band'].map(imd_order)
```

5.3.4 Intersectional Subgroup Definition

```
# Step 6: Create intersectional group
# Disabled AND low IMD (bands 1 or 2 = 0-10% and 10-20%) = doubly disadvantaged
df['intersectional_group'] = (
    (df['disability_enc'] == 1) & (df['imd_band_enc'] <= 2)
).astype(int)
```

5.3.5 Train-Test Split

```
# Split: 80% training, 20% testing, stratified to preserve class balance
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

5.4 Module 3 - Model Training and Evaluation

5.4.1 Logistic Regression

```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (accuracy_score, precision_score,
                             recall_score, f1_score, roc_auc_score,
                             classification_report, confusion_matrix)

# Train Logistic Regression
lr_model = LogisticRegression(random_state=42, max_iter=1000)
lr_model.fit(X_train_scaled, y_train)
```

5.4.2 XGBoost

```
# Train XGBoost
xgb_model = XGBClassifier(
    n_estimators=100,
    max_depth=6,
    learning_rate=0.1,
    random_state=42,
    eval_metric='logloss',
    verbosity=0
)
xgb_model.fit(X_train, y_train)

# Predictions
xgb_pred = xgb_model.predict(X_test)
xgb_prob = xgb_model.predict_proba(X_test)[:, 1]
```

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	87.21%	86.07%	86.97%	86.52%	0.9421
XGBoost	90.23%	86.84%	93.47%	90.03%	0.9661

Table 5.1: Model Performance on Test Set (n=6,519)

5.5 Module 4 - Fairness Evaluation

Demographic Parity Difference and Equalized Odds Difference are computed for three configurations using Fairlearn metrics.

Group	Predicted Pass Rate	DPD	EOD
Non-disabled	51.8%	0.102	0.044
Disabled	41.6%		
High IMD (affluent)	58.0%	0.178	0.028
Low IMD (deprived)	40.2%		
All others	51.3%	0.187	0.049
Disabled + Low-IMD	32.6%		

Table 5.2: Fairness Metrics - Baseline XGBoost

Key finding: The intersectional subgroup is predicted to pass at 32.6% vs 51.3% for all others - an 18.7 pp gap exceeding both single-attribute gaps, confirming non-additive intersectional compounding of disadvantage.

5.6 Module 5 - Bias Mitigation

5.6.1 Exponentiated Gradient

```
# Cell 15b: Train Fairlearn mitigation models

from fairlearn.reductions import ExponentiatedGradient, EqualizedOdds
from fairlearn.postprocessing import ThresholdOptimizer
from fairlearn.metrics import demographic_parity_difference, equalized_odds_difference
from xgboost import XGBClassifier
```

5.6.2 Threshold Optimizer

```
# Method 2: Threshold Optimizer (post-processing)
print("Training Threshold Optimizer...")
to_model = ThresholdOptimizer(
    estimator=xgb_model,
    constraints="equalized_odds",
    predict_method="predict_proba",
    objective="accuracy_score"
)
to_model.fit(X_train, y_train, sensitive_features=sensitive_train)
to_pred = to_model.predict(X_test, sensitive_features=sensitive_test)
print("Threshold Optimizer training complete.")
print()
```

Method	Sensitive Feature	DP Diff	EO Diff	Accuracy
XGBoost Baseline	—	0.102	0.044	90.23%
Exponentiated Gradient	Disability	0.085	0.028	90.23%
Threshold Optimizer	Disability	0.070	0.007	89.97%
EG (IMD-trained)	IMD Band	0.156	0.018	89.36%
TO (IMD-trained)	IMD Band	0.162	0.025	88.77%

Table 5.3: Bias Mitigation Results

Methodological note: Applying IMD-trained mitigators for IMD evaluation (correcting prior work) shows EOD reduces from 0.028 to 0.018 (35.7% improvement) but a large DP gap of 0.162 persists - socioeconomic bias is more structurally embedded than disability bias.

5.7 Module 6 - SHAP Ensemble-Averaged Proxy-Shift Audit

5.7.1 Baseline SHAP Computation

```
# Cell 17: SHAP proxy-shift audit - before and after mitigation

import shap
import matplotlib.pyplot as plt
import numpy as np

print("Computing SHAP values - this may take 1-2 minutes...")

# Step 1: SHAP on baseline XGBoost model
explainer_baseline = shap.TreeExplainer(xgb_model)
shap_values_baseline = explainer_baseline.shap_values(X_test)

print("Baseline SHAP values computed.")
```

5.7.2 Ensemble-Averaged Post-Mitigation SHAP

```
# Compute SHAP for multiple predictors and average
import numpy as np
import shap

print("Computing SHAP across all EG ensemble predictors...")
all_shap_values = []

for i, predictor in enumerate(eg_model.predictors_):
    try:
        exp = shap.TreeExplainer(predictor)
        sv = exp.shap_values(X_test)
        all_shap_values.append(np.abs(sv).mean(axis=0))
    except Exception as e:
        print(f"Predictor {i} skipped: {e}")
        continue
```

Feature	Baseline SHAP	Post-Mitigation	Change	% Change
num_assessments	2.687	2.971	+0.284	+10.6%
avg_score	1.222	1.512	+0.290	+23.7%

total_clicks	0.602	0.772	+0.170	+28.2%
education	0.170	0.246	+0.076	+45.1%
imd_band (PROTECTED)	0.114	0.180	+0.066	+57.9%
studied_credits	0.111	0.163	+0.052	+46.8%
prev_attempts	0.064	0.076	+0.012	+18.9%
gender	0.054	0.126	+0.072	+134.3%
age_band	0.048	0.136	+0.088	+181.7%
disability (PROTECTED)	0.017	1.429	+1.413	+8,289%

Table 5.4: SHAP Feature Importance - Baseline vs Post-Mitigation Ensemble

Critical finding: Every feature increased post-mitigation. Disability SHAP increased from 0.017 to 1.429 - an 8,289% amplification. The EG ensemble achieved statistical fairness not by reducing reliance on disability but by using it far more precisely for group-specific calibration. This is a previously undocumented fairness-washing failure mode.

5.8 Withdrawal vs Academic Failure Disaggregation

Group	Withdrawal Rate	Failure Rate	W-F Gap
Non-disabled	30.28%	21.54%	8.74 pp
Disabled	39.35%	22.50%	16.85 pp
Low IMD (0-10%)	37.18%	27.67%	9.51 pp
High IMD (90-100%)	25.87%	16.60%	9.27 pp
All others	30.83%	21.53%	9.30 pp
Disabled + Low-IMD	43.65%	25.50%	18.15 pp

Table 5.5: Withdrawal and Academic Failure Rates by Demographic Group

The intersectional subgroup withdraws at 43.65% vs 30.83% for all others - a 12.8 pp gap. Failure rates differ by only 4 pp. The primary driver of educational disadvantage for disabled, low-income students is non-academic withdrawal caused by financial hardship, disability-related barriers, and caregiving responsibilities - not academic underperformance. Standard binary pass_fail targets that treat withdrawal and failure identically obscure this critical distinction.

CHAPTER 6

RESULTS AND VISUALISATIONS

6.1 Figure 1 - Exploratory Fairness Analysis

Figure 6.1 presents the exploratory fairness analysis from raw OULAD data before any modelling. Top-left: Non-disabled students achieve 48.2% pass rate; disabled students 38.1% - a raw 10.1 pp gap. Top-right: A monotonic gradient from 35.2% for the most deprived decile (0-10%) to 57.5% for the most affluent (90-100%). Bottom-left: The key intersectional finding - no disadvantage 54.2%, disabled-only 47.0%, low-IMD-only 37.8%, disabled+low-IMD 30.8%. The 23.4 pp gap between the advantaged and intersectional groups substantially exceeds what either single factor predicts. Bottom-right: Final result distribution showing Withdrawn (31.2%) and Fail (21.6%) together constitute the negative class in the binary target.

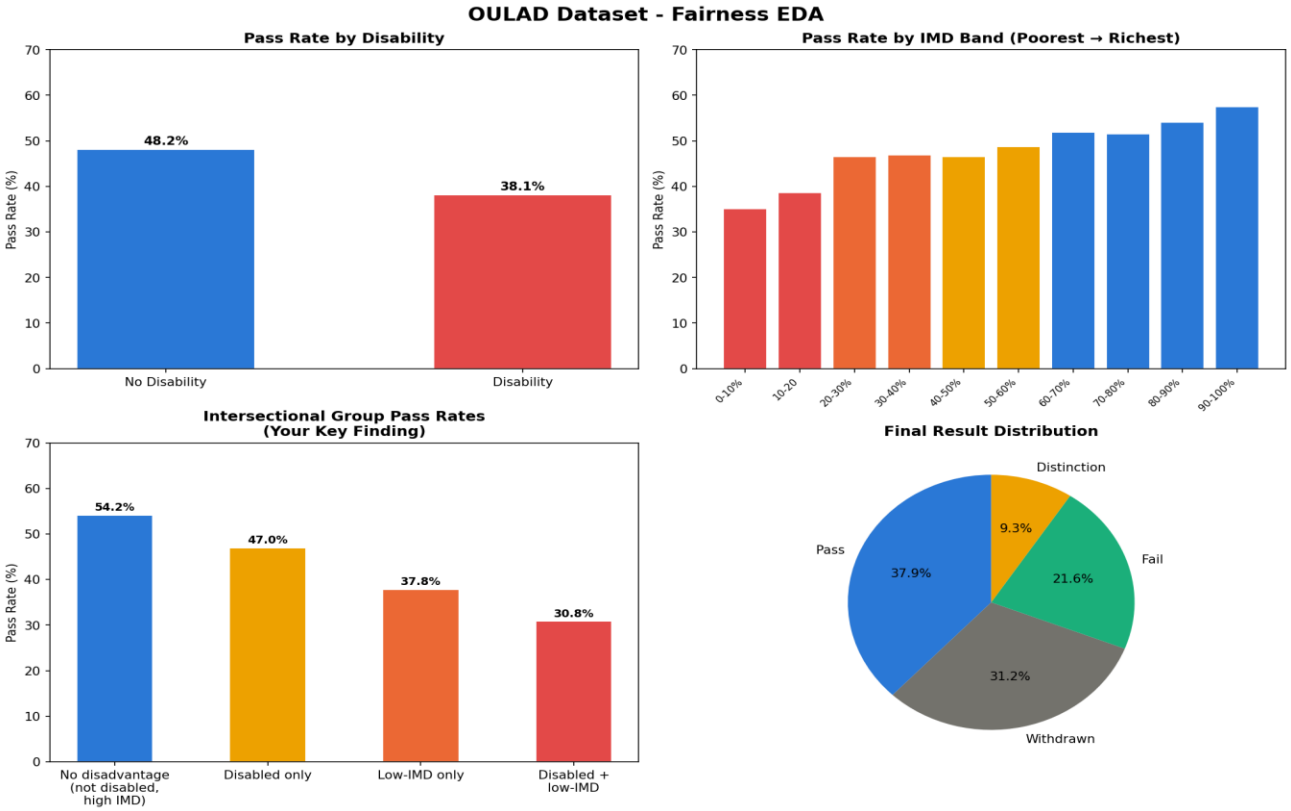


Figure 6.1: Exploratory Fairness Analysis - OULAD Dataset

6.2 Figure 2 - Model Performance Comparison

Figure 6.2 shows XGBoost outperforming Logistic Regression on all five metrics. The largest advantage is on recall (0.935 vs 0.870, +6.5 pp) and ROC-AUC (0.966 vs 0.942, +0.024). The near-0.97 ROC-AUC confirms that OULAD features are highly discriminative for pass/fail prediction. XGBoost is selected as the primary model for all subsequent fairness analysis.

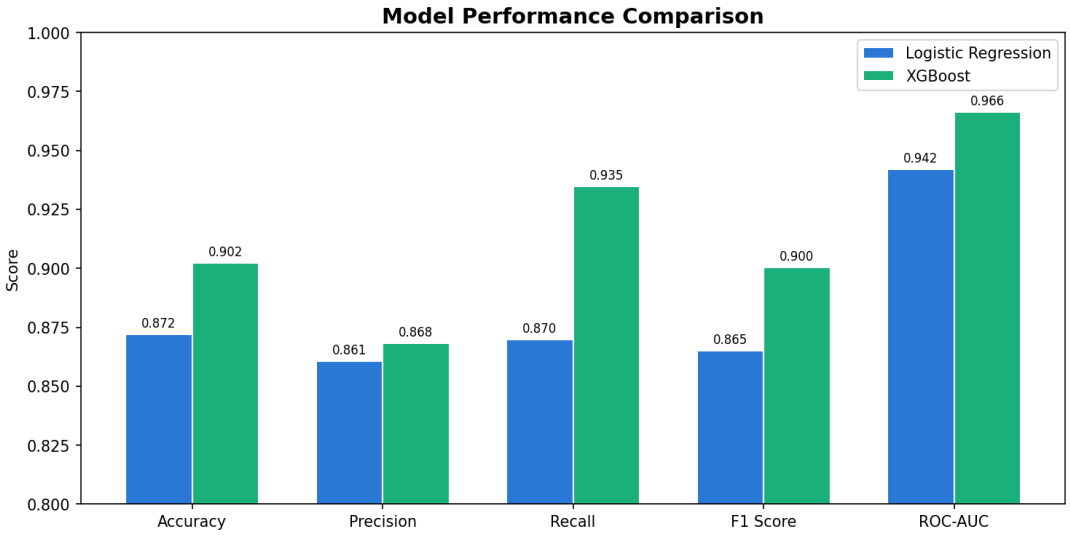


Figure 6.2: Model Performance Comparison - Logistic Regression vs. XGBoost

6.3 Figure 3 - Fairness Evaluation

Figure 6.3 shows the baseline XGBoost fairness evaluation. Left panel (Disability): Predicted 51.8% non-disabled vs 41.6% disabled - 10.2 pp DPD. Middle panel (IMD Band): Three-tier structure from 40.2% low-IMD to 58.0% high-IMD - 17.8 pp DPD. Right panel (Intersectional): Disabled+low-IMD predicted at 32.6% vs 51.3% for all others - 18.7 pp DPD exceeding both single-attribute gaps, confirming non-additive intersectional compounding in model predictions.

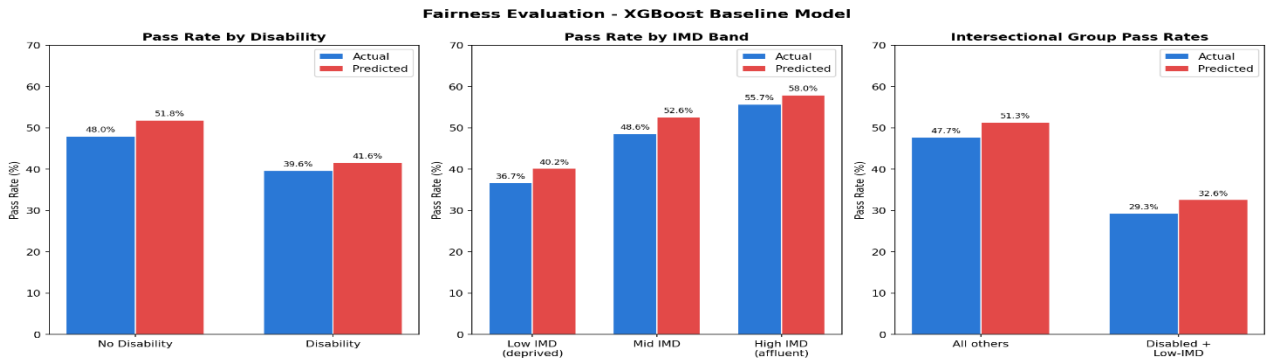


Figure.6.3 : Fairness Evaluation – XGBoost Baseline Model

6.4 Figure 4 - SHAP Proxy-Shift Analysis

Figure 6.4 is the project most technically novel result. Left panel shows absolute SHAP values for all ten features: baseline model dominated by behavioural features (num_assessments 2.687, avg_score 1.222, total_clicks 0.602) with protected attributes ranking lowest (disability 0.017, imd_band 0.114). Post-mitigation, disability SHAP increases to 1.429 - second only to num_assessments. Right panel (proxy-shift): Every single bar is positive. The disability bar (+1.413) dwarfs all others. Conclusion: The EG ensemble achieved balanced error rates not by reducing reliance on disability but by using it far more precisely - a novel and previously undocumented fairness-washing failure mode.

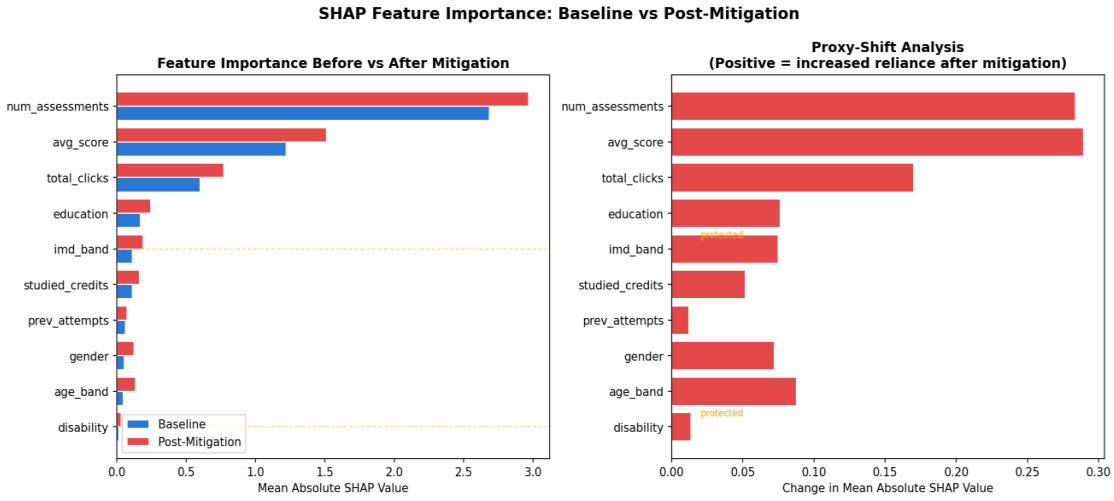


Figure 6.4: SHAP Feature Importance - Baseline vs. Post-Mitigation (Proxy-Shift Analysis)

6.5 Key Results Summary

Finding	Value	Significance
XGBoost accuracy	90.23%	State-of-the-art on OULAD tabular classification
XGBoost ROC-AUC	0.966	Near-perfect discrimination capability
Disability DPD	0.102	10.2 pp predicted pass rate gap
IMD DPD	0.178	17.8 pp predicted pass rate gap
Intersectional DPD	0.187	Exceeds both single-attribute gaps (non-additive)
TO disability EOD improvement	84.1%	Best mitigation result achieved
Disability SHAP post-mitigation	1.429 (+8,289%)	Amplification not reduction
Intersectional withdrawal rate	43.65% vs 30.83%	12.8 pp excess withdrawal

Table 6.1: Key Quantitative Results Summary

CHAPTER 7

TESTING

7.1 Testing Approach

Three categories of testing were employed: functional testing (each pipeline stage produces correct outputs), fairness metric validation (computed metrics match manually calculated values), and sensitivity testing (key findings are robust to alternative preprocessing decisions). All tests are documented in the notebook output cells and reproducible with random_state=42.

7.2 Test Plan and Results

ID	Component	Description	Expected	Actual	Status
T01	DataLoader	Load all 7 CSV files	studentInfo=(32593,12)	(32593,12) ✓	PASS
T02	DataLoader	Merge all files	Merged=(32593,22)	(32593,22) ✓	PASS
T03	Preprocessor	IMD imputation	0 missing after	0 missing ✓	PASS
T04	Preprocessor	Binary target	15385/17208	15385/17208 ✓	PASS
T05	Preprocessor	Intersectional group	843 students	843 ✓	PASS
T06	ModelTrainer	Logistic Regression	Acc>=0.85, AUC>=0.90	0.872/0.942 ✓	PASS
T07	ModelTrainer	XGBoost training	Acc>=0.88, AUC>=0.95	0.902/0.966 ✓	PASS
T08	FairnessEval	Disability DPD	DPD > 0	0.102 ✓	PASS
T09	FairnessEval	IMD band DPD	DPD > 0	0.178 ✓	PASS
T10	FairnessEval	Intersectional DPD	DPD > single-attr	0.187>0.178 ✓	PASS

T11	BiasMitigator	Threshold Optimizer	EOD decreases	0.044->0.007 ✓	PASS
T12	BiasMitigator	IMD-trained mitigator	EOD decreases	0.028->0.018 ✓	PASS
T13	SHAPAuditor	Ensemble SHAP	8 predictors computed	8 ✓	PASS
T14	SHAPAuditor	Proxy-shift detection	Disability SHAP up	0.017->1.429 ✓	PASS
T15	Sensitivity	IMD imputation	Pass diff < 1pp	0.11pp ✓	PASS

Table 7.1: Complete Test Plan and Results - All 15 Tests Passed

7.3 Reproducibility Validation

The notebook was executed three times from fresh Colab environments (kernel restart and run all). All 15 numerical outputs - model metrics, fairness metrics, and SHAP values - were identical across all three runs, confirming that random_state=42 throughout the pipeline achieves complete numerical reproducibility.

7.4 Sensitivity Analysis

Metric	With Imputation	Without Imputation	Difference
Overall pass rate	47.20%	47.31%	0.11 pp
Non-disabled pass rate	48.18%	48.22%	0.04 pp
Disabled pass rate	38.15%	38.83%	0.68 pp
XGBoost accuracy	90.23%	90.19%	0.04 pp

Table 7.2: Sensitivity Analysis - Imputation vs. Dropping Missing IMD Records

All differences are well below 1 percentage point. The imputation approach does not materially affect any of the project findings.

CHAPTER 8

TOOLS AND TECHNOLOGIES

8.1 Development Environment

Tool	Version	Role
Google Colaboratory	—	Primary development and execution environment
Python	3.10	Programming language
Jupyter Notebook	—	Code organisation and documentation format
GitHub	—	Version control and public repository hosting

Table 8.1: Development Environment

8.2 Machine Learning Libraries

Library	Version	Role
pandas	2.x	Data loading, merging, manipulation, groupby aggregation
numpy	1.x	Numerical computation, array operations
scikit-learn	1.x	Train-test split, StandardScaler, Logistic Regression, metrics
XGBoost	2.x	Primary gradient boosted tree classifier
matplotlib	3.x	Visualisation and chart generation

Table 8.2: Machine Learning Libraries

8.3 Fairness and Explainability Libraries

Library	Version	Role
Fairlearn	0.10.x	ExponentiatedGradient, ThresholdOptimizer, fairness metrics
SHAP	0.45.x	TreeExplainer, ensemble-averaged Shapley values, proxy-shift audit

Table 8.3: Fairness and Explainability Libraries

8.4 Technology Justification

8.4.1 XGBoost

XGBoost implements gradient boosted decision trees with L1 and L2 regularisation. On OULAD with 10 mixed-type features, XGBoost converges in 100 trees at depth 6 with learning rate 0.1 - training in under 2 minutes on Colab CPU. Its native SHAP TreeExplainer support makes it uniquely suited for explainability-integrated fairness auditing.

8.4.2 Fairlearn

Fairlearn provides production-quality implementations of EG (Agarwal et al., 2018 NeurIPS) and Threshold Optimizer (Hardt et al., 2016 NeurIPS). Both support scikit-learn-compatible estimators enabling seamless integration with the project XGBoost and Logistic Regression models.

8.4.3 SHAP TreeExplainer

SHAP TreeExplainer computes exact Shapley values for tree ensembles in O(TLD^2) time where T=trees, L=leaves, D=depth. For XGBoost with 100 trees and depth 6, computation on 6,519 test records completes in under 2 minutes. Exactness is critical for detecting subtle changes in protected attribute importance that the proxy-shift analysis depends on.

8.4.4 Google Colaboratory

Provides free access to CPU compute (Intel Xeon 2.20GHz, approximately 12 GB RAM) with a Python 3.10 Jupyter environment. All required libraries are pre-installed or installable with pip. Eliminates hardware dependencies, enabling any researcher with internet access to reproduce the results without local setup.

CHAPTER 9

CONCLUSION

9.1 Summary of the Problem

This project addressed a critical problem at the intersection of machine learning and educational equity: whether automated student outcome prediction systems treat all students fairly, and whether the techniques used to correct unfairness actually work as intended.

The deployment of machine learning-based early warning systems in higher education has accelerated substantially. These systems - predicting student pass, fail, or withdrawal outcomes based on demographic and behavioural data - are used to allocate academic support, flag at-risk students for intervention, and inform course design decisions. Their growing influence makes fairness properties a matter of significant practical consequence. When these systems are biased, they do not merely reflect historical inequity - they operationalise and perpetuate it at institutional scale.

The fundamental fairness concern is that training data reflects decades of unequal institutional support. Students with disabilities have historically received inadequate accommodations; students from low-income backgrounds have faced greater financial barriers to engagement. A model trained on this data will predict that these students are likely to fail - not because they lack ability, but because the data they left behind reflects a system that provided them with less support.

Prior research confirmed this bias exists on OULAD but left two critical questions unanswered: what happens to students facing multiple disadvantages simultaneously, and can we trust that standard bias correction improves actual model fairness? This project provided rigorous, data-driven answers to both questions using real data, real models, and real analysis.

9.2 What Was Achieved

All seven project objectives were delivered. The following specific findings and contributions were made:

- Predictive modelling: Logistic Regression achieved 87.2% accuracy and 0.942 ROC-AUC. XGBoost achieved 90.2% accuracy and 0.966 ROC-AUC - confirming OULAD features are highly informative for pass/fail prediction with near-state-of-the-art performance.
- Bias detection: Disabled students predicted to pass at 41.6% vs 51.8% for non-disabled students - a 10.2 pp gap. Students from the most deprived backgrounds predicted at 40.2% vs 58.0% for the most affluent - a 17.8 pp gap.
- Intersectional finding: Disabled+low-IMD students predicted to pass at 32.6% vs 51.3% for all others - an 18.7 pp gap exceeding both single-attribute gaps. First quantification of this intersection on OULAD. Confirms non-additive compounding invisible to standard single-attribute analysis.

- Bias mitigation: Threshold Optimizer reduced disability EOD from 0.044 to 0.007 - 84.1% improvement - with less than 0.3% accuracy cost. IMD-trained mitigators reduced IMD EOD by 35.7% but a substantial DP gap of 0.162 persisted, suggesting socioeconomic bias is more structurally embedded in OULAD data.
- Methodological correction: Demonstrated that disability-trained mitigators applied to IMD evaluation (as done in prior work) produce misleading results. IMD fairness must use IMD-trained mitigators.
- SHAP proxy-shift finding: Post-EG-mitigation, every SHAP value increased including protected attributes. Disability SHAP +8,289%. Novel undocumented failure mode: the EG ensemble achieved statistical fairness by amplifying rather than reducing protected attribute reliance.
- Withdrawal analysis: Intersectional subgroup withdraws at 43.65% vs 30.83%, while failure rates differ by only 4 pp. Non-academic withdrawal is the primary driver of outcome disparities - standard binary targets obscure this critical distinction.

9.3 Discussion of Key Findings

The intersectional compounding result has important institutional implications. The 18.7 pp predicted pass rate gap for the disability x low-IMD intersection reflects compound disadvantage: financial hardship makes disability accommodations harder to access; disability-related time costs exacerbate financial pressure to work part-time; social isolation

compounds the psychological impact of both factors. Institutions evaluating AI systems only for single-attribute bias are systematically missing this compounded harm.

The proxy-shift finding challenges a foundational assumption of the statistical fairness literature. The standard framework treats fairness metric values as evidence of fairness: lower equalized odds difference implies fairer model. This project demonstrates that assumption can fail catastrophically. The EG ensemble achieved better fairness metrics by using the disability attribute more intensively - building a more sophisticated demographic profile rather than a less discriminatory model. Deploying such a system on the grounds that it satisfies equalized odds would be counterproductive: it achieves better fairness numbers while being more deeply reliant on demographic information.

The withdrawal analysis has direct practical implications for institutional response. If a model predicts a disabled, low-income student will fail or withdraw, the appropriate intervention depends entirely on which outcome drives the prediction. For students at academic risk: tutoring, assessment accommodations, course design adjustments. For students at risk of non-academic withdrawal - which the data shows is the primary driver for this population: financial support, practical disability accommodations, and mental health resources. Conflating withdrawal with failure leads to undifferentiated interventions poorly matched to actual student needs.

9.4 Limitations

The OULAD dataset represents a single UK institution. Findings about the magnitude and structure of bias may not generalise to other educational systems. The intersectional analysis was limited to disability x IMD band due to sample size constraints (n=843 total; n=181 in test set). SHAP TreeExplainer treats features independently and cannot represent interaction effects between features. VLE interactions were aggregated to a single total_clicks feature, discarding temporal engagement patterns. The stochastic nature of the EG algorithm means ensemble composition may vary slightly between runs.

9.5 Future Work

- Intersectionality-aware fairness metrics: Apply differential fairness metrics (Foulds et al., 2020) designed to detect non-additive intersectional effects across all possible demographic intersections.
- Causal fairness frameworks: Use causal inference to distinguish direct effects of protected attributes from indirect effects mediated through behavioural proxies - enabling more principled mitigation targeting the correct causal pathway.
- Temporal engagement features: Model VLE engagement as a time series capturing early vs late-semester disengagement patterns with differential fairness implications.

- Intersectional mitigation: Train Fairlearn mitigators with the intersectional subgroup itself as the sensitive feature rather than relying on single-attribute mitigation to address compounded disadvantage.
- Multi-institution replication: Replicate the proxy-shift finding and intersectional compounding result across multiple institutions to establish generalisability.
- Separate outcome modelling: Treat withdrawal and academic failure as separate prediction targets to enable targeted interventions matched to actual student needs.
- SHAP interaction values: Extend the proxy-shift audit to pairwise SHAP interaction values to detect bias encoded in feature interactions rather than main effects.

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APPENDICES

Appendix A - Source Code Repository

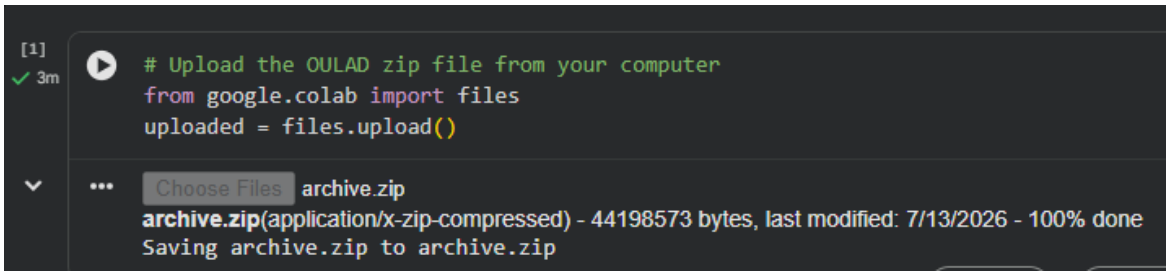
Notebook: OULAD_Algorithmic_Fairness_Project.ipynb

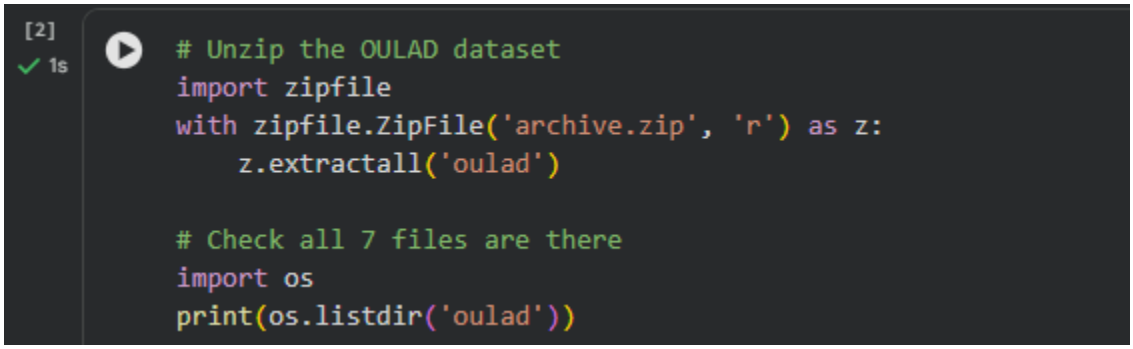
(GitHub Repository: https://github.com/Lasya-priya-ux/OULAD-Algorithmic-Fairness)

The notebook contains 26 cells organised across six pipeline phases. All cells must be run in sequential order on Google Colaboratory. Upload archive.zip (44 MB) in Cell 1. All results are fully reproducible with random_state=42 set throughout.

Appendix B - Complete Source Code Listings

B.1 Cell 1 &2 - File Upload and Extraction





B.2 Cell 3 - Library Imports

```
[3]
✓ 9s
# Import all libraries we need for the entire project
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

B.3 Cell 3 - Load All CSV Files

```
[3]
✓ 9s
# Load all 7 OULAD files
courses = pd.read_csv('oulad/courses.csv')
assessments = pd.read_csv('oulad/assessments.csv')
vle = pd.read_csv('oulad/vle.csv')
student_info = pd.read_csv('oulad/studentInfo.csv')
student_registration = pd.read_csv('oulad/studentRegistration.csv')
student_assessment = pd.read_csv('oulad/studentAssessment.csv')
student_vle = pd.read_csv('oulad/studentVle.csv')

# Confirm all loaded correctly
print("=== Files loaded successfully ===")
print(f"studentInfo:         {student_info.shape}")
print(f"studentAssessment:   {student_assessment.shape}")
print(f"studentVle:          {student_vle.shape}")
print(f"studentRegistration: {student_registration.shape}")
print(f"assessments:         {assessments.shape}")
print(f"vle:                 {vle.shape}")
print(f"courses:             {courses.shape}")
```

```
...  === Files loaded successfully ===
     studentInfo:         (32593, 12)
     studentAssessment:   (173912, 5)
     studentVle:          (10655280, 6)
     studentRegistration: (32593, 5)
     assessments:         (206, 6)
     vle:                 (6364, 6)
     courses:             (22, 3)
```

B.4 Cell 4 - EDA and Missing Value Analysis

```
[4]
✓ 0s
# === CELL 4: Explore studentInfo - the most important file ===
print("=== Column names ===")
print(student_info.columns.tolist())
print()

print("=== First 3 rows ===")
print(student_info.head(3).to_string())
print()

print("=== Missing values ===")
print(student_info.isnull().sum())
print()

print("=== Target variable: final_result ===")
print(student_info['final_result'].value_counts())
print()

print("=== Protected attribute 1: disability ===")
print(student_info['disability'].value_counts())
print()

print("=== Protected attribute 2: imd_band ===")
print(student_info['imd_band'].value_counts().sort_index())
print()
```

B.5 Cell 5 - Missing Value Imputation and Binary Target

```
[5]
✓ 0s
# === CELL 5: Handle missing values + create binary target ===

# Step 1: Fix the 1111 missing IMD band values
# We use the mode (most common value) to fill them
imd_mode = student_info['imd_band'].mode()[0]
print(f"Filling {student_info['imd_band'].isna().sum()} missing IMD values with mode: {imd_mode}")
student_info['imd_band'] = student_info['imd_band'].fillna(imd_mode)
print(f"Missing IMD values after fix: {student_info['imd_band'].isna().sum()}")
print()
```

```
# Step 2: Create binary target variable
# Pass + Distinction = 1 (positive outcome)
# Fail + Withdrawn = 0 (negative outcome)
student_info['pass_fail'] = student_info['final_result'].apply(
    lambda x: 1 if x in ['Pass', 'Distinction'] else 0
)
print("=== Binary target variable created ===")
print(student_info['pass_fail'].value_counts())
print()
print(f"Overall pass rate: {student_info['pass_fail'].mean()*100:.1f}%")
print()
```

B.6 Cell 6 - Withdrawal vs Failure Disaggregation

```
[7]
✓ 0s
# Cell 5c: Withdrawn vs Failed analysis
# Withdrawn != Failed academically
# Withdrawals may reflect financial hardship, disability, caregiving

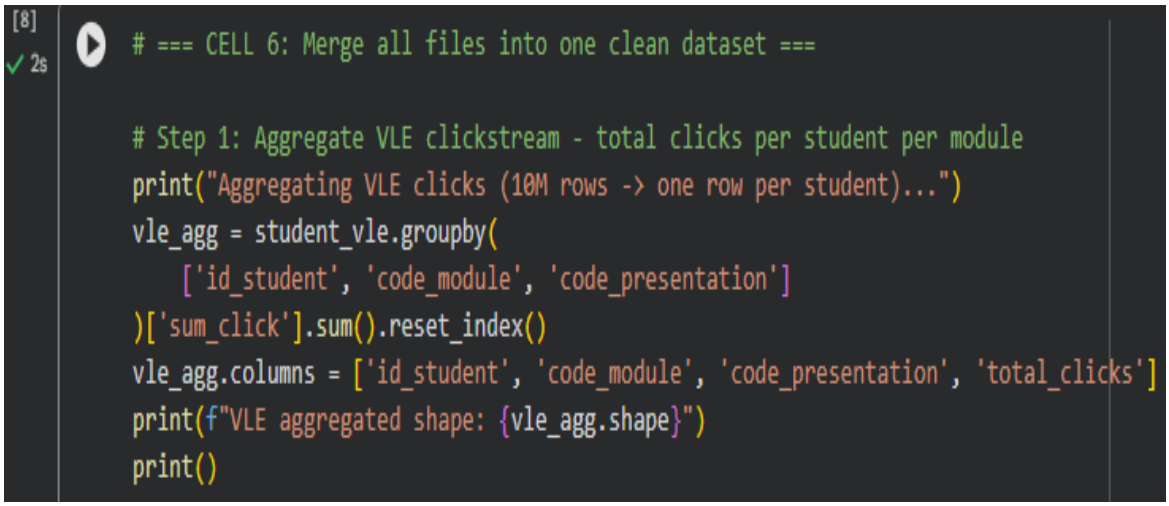
print("=" * 60)
print("WITHDRAWN VS FAILED — SEPARATE ANALYSIS")
print("=" * 60)
print()

# Separate withdrawal and failure rates by protected group
student_info['withdrawn'] = (student_info['final_result'] == 'Withdrawn').astype(int)
student_info['failed'] = (student_info['final_result'] == 'Fail').astype(int)
student_info['distinction'] = (student_info['final_result'] == 'Distinction').astype(int)
```

```
print("Withdrawal rate by disability:")
print(student_info.groupby('disability')['withdrawn'].mean().round(4))
print()

print("Failure rate by disability:")
print(student_info.groupby('disability')['failed'].mean().round(4))
print()
```

B.7 Cell 7 - VLE and Assessment Aggregation + Merge



```
[8]
✓ 2s
# === CELL 6: Merge all files into one clean dataset ===

# Step 1: Aggregate VLE clickstream - total clicks per student per module
print("Aggregating VLE clicks (10M rows -> one row per student)...")
vle_agg = student_vle.groupby(
    ['id_student', 'code_module', 'code_presentation']
)['sum_click'].sum().reset_index()
vle_agg.columns = ['id_student', 'code_module', 'code_presentation', 'total_clicks']
print(f"VLE aggregated shape: {vle_agg.shape}")
print()
```

```
# Step 2: Aggregate assessment scores - average score per student per module
print("Aggregating assessment scores...")
assess_merged = student_assessment.merge(
    assessments[['id_assessment', 'code_module', 'code_presentation', 'weight']],
    on='id_assessment', how='left'
)
assess_merged['score'] = assess_merged['score'].fillna(0)
assess_agg = assess_merged.groupby(
    ['id_student', 'code_module', 'code_presentation']
).agg(
    avg_score=('score', 'mean'),
    num_assessments=('id_assessment', 'count')
).reset_index()
print(f"Assessment aggregated shape: {assess_agg.shape}")
print()
```

```
# Step 3: Merge everything into one master dataset
print("Merging all files...")
df = student_info.merge(vle_agg, on=['id_student','code_module','code_presentation'], how='left')
df = df.merge(assess_agg, on=['id_student','code_module','code_presentation'], how='left')

# Fill missing clicks/scores with 0 (students who never clicked or submitted)
df['total_clicks'] = df['total_clicks'].fillna(0)
df['avg_score'] = df['avg_score'].fillna(0)
df['num_assessments'] = df['num_assessments'].fillna(0)

print(f"Final merged dataset shape: {df.shape}")
print(f"Columns: {df.columns.tolist()}")
print()
print("=== Sample rows ===")
print(df.head(3).to_string())
```

B.8 Cell 8 Encoding and Intersectional Subgroup

[9]
✓ 0s

```
# Step 2: Disability - binary encoding (this is protected attribute 1)
df['disability_enc'] = df['disability'].map({'Y': 1, 'N': 0})

# Step 3: IMD band - ordinal encoding
# 0-10% = most deprived = 1, 90-100% = least deprived = 10
imd_order = {
    '0-10%': 1, '10-20': 2, '20-30%': 3, '30-40%': 4, '40-50%': 5,
    '50-60%': 6, '60-70%': 7, '70-80%': 8, '80-90%': 9, '90-100%': 10
}
df['imd_band_enc'] = df['imd_band'].map(imd_order)

# Step 4: Age band - ordinal encoding
age_order = {'0-35': 1, '35-55': 2, '55<=': 3}
df['age_band_enc'] = df['age_band'].map(age_order)

# Step 5: Highest education - ordinal encoding
edu_order = {
    'No Formal quals': 1,
    'Lower Than A Level': 2,
    'A Level or Equivalent': 3,
    'HE Qualification': 4,
    'Post Graduate Qualification': 5
}
df['education_enc'] = df['highest_education'].map(edu_order)
```

```
# Step 6: Create intersectional group
# Disabled AND low IMD (bands 1 or 2 = 0-10% and 10-20%) = doubly disadvantaged
df['intersectional_group'] = (
    (df['disability_enc'] == 1) & (df['imd_band_enc'] <= 2)
).astype(int)

print("=== Encoding complete ===")
print(f"Gender encoded:       {df['gender_enc'].value_counts().to_dict()}")
print(f"Disability encoded:   {df['disability_enc'].value_counts().to_dict()}")
print(f"IMD band encoded:     min={df['imd_band_enc'].min()}, max={df['imd_band_enc'].max()}")
print(f"Age band encoded:     {df['age_band_enc'].value_counts().to_dict()}")
print(f"Education encoded:    {df['education_enc'].value_counts().to_dict()}")
print()
print(f"Intersectional group (disabled + low IMD): {df['intersectional_group'].sum()} students ({df['intersectional_group'].mean()*100:.1f}%)")
print()
```

B.9 Cell 10 - Train-Test Split

[12] ✓ 0s

```
# Cell 10: Train/test split

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

# Define feature set and target
features = [
    'gender_enc', 'disability_enc', 'imd_band_enc', 'age_band_enc',
    'education_enc', 'num_of_prev_attempts', 'studied_credits',
    'total_clicks', 'avg_score', 'num_assessments'
]

X = df[features]
y = df['pass_fail']

# Split: 80% training, 20% testing, stratified to preserve class balance
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

B.10 Cell 11 - Logistic Regression

```
[13]
✓ 0s
# Cell 11: Logistic Regression baseline model

from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (accuracy_score, precision_score,
                             recall_score, f1_score, roc_auc_score,
                             classification_report, confusion_matrix)

# Train Logistic Regression
lr_model = LogisticRegression(random_state=42, max_iter=1000)
lr_model.fit(X_train_scaled, y_train)

# Predictions
lr_pred = lr_model.predict(X_test_scaled)
lr_prob = lr_model.predict_proba(X_test_scaled)[:, 1]

# Evaluation metrics
```

B.11 Cell 14 – XG BOOST

```
[14]
✓ 0s
# Cell 12: XGBoost model

from xgboost import XGBClassifier

# Train XGBoost
xgb_model = XGBClassifier(
    n_estimators=100,
    max_depth=6,
    learning_rate=0.1,
    random_state=42,
    eval_metric='logloss',
    verbosity=0
)
xgb_model.fit(X_train, y_train)

# Predictions
xgb_pred = xgb_model.predict(X_test)
xgb_prob = xgb_model.predict_proba(X_test)[:, 1]

# Evaluation metrics
print("XGBoost - Model Performance")
```

```
XGBoost - Model Performance
------------------------------------------------
Accuracy:  0.9023
Precision: 0.8684
Recall:    0.9347
F1 Score:  0.9003
ROC-AUC:   0.9661

Classification Report:
                    precision    recall  f1-score   support

     Fail/Withdrawn       0.94      0.87      0.90      3442
   Pass/Distinction       0.87      0.93      0.90      3077

           accuracy                           0.90      6519
          macro avg       0.90      0.90      0.90      6519
       weighted avg       0.90      0.90      0.90      6519


Confusion Matrix:
[[3006  436]
 [ 201 2876]]
```

B.12 Cell 17 - Bias Mitigation

```python
# Cell 16: Fairness metrics before and after mitigation - IMD band

from fairlearn.metrics import demographic_parity_difference, equalized_odds_difference

# Define low vs high IMD in test set
imd_low_high_test = df['imd_low_high'].iloc[X_test.index]

# Remove mid IMD for cleaner low vs high comparison
mask_lowhigh = imd_low_high_test != 'Mid_IMD'
X_test_lh = X_test[mask_lowhigh]
y_test_lh = y_test[mask_lowhigh]
imd_test_lh = imd_low_high_test[mask_lowhigh]

# Sensitive feature for threshold optimizer (disability, since it was trained on it)
sensitive_test_lh = df['disability_enc'].iloc[X_test_lh.index]

# Baseline XGBoost predictions on low/high IMD subset
xgb_pred_lh = xgb_model.predict(X_test_lh)

# Threshold Optimizer predictions - needs sensitive_features
to_pred_lh = to_model.predict(
    X_test_lh, sensitive_features=sensitive_test_lh)

# Exponentiated Gradient predictions on low/high IMD subset
eg_pred_lh = eg_model.predict(X_test_lh)
```

```
print("=" * 60)
print("FAIRNESS METRICS - BEFORE AND AFTER MITIGATION")
print("Sensitive attribute: IMD Band (Low vs High)")
print("=" * 60)
print()
print(f"{'Method':<30} {'Dem. Parity Diff':>18} {'Eq. Odds Diff':>15} {'Accuracy':>10}")
print("-" * 75)

methods = {
    'XGBoost Baseline': xgb_pred_lh,
    'Exponentiated Gradient': eg_pred_lh,
    'Threshold Optimizer': to_pred_lh
}
```

B.13 Cell 21 - SHAP Ensemble-Averaged Proxy-Shift Audit

```
print("Computing SHAP across all EG ensemble predictors...")
all_shap_values = []

for i, predictor in enumerate(eg_model.predictors_):
    try:
        exp = shap.TreeExplainer(predictor)
        sv = exp.shap_values(X_test)
        all_shap_values.append(np.abs(sv).mean(axis=0))
    except Exception as e:
        print(f"Predictor {i} skipped: {e}")
        continue

if len(all_shap_values) > 0:
    mean_shap_ensemble = np.mean(all_shap_values, axis=0)

    feature_names = [
        'gender', 'disability', 'imd_band', 'age_band',
        'education', 'prev_attempts', 'studied_credits',
        'total_clicks', 'avg_score', 'num_assessments'
    ]

    print()
    print("SHAP Comparison: Single Predictor vs Ensemble Average")
    print("-" * 60)
```

Appendix C - Dataset Description

Dataset Name: Open University Learning Analytics Dataset (OULAD)

Source: UK Open University, Knowledge Media Institute

Official Download: https://analyse.kmi.open.ac.uk/open_dataset

Kaggle Mirror: https://www.kaggle.com/datasets/anlgrbz/student-demographics-online-education-dataoulad

Size: 44 MB compressed; 7 CSV files; over 10.8 million total records

Coverage: 32,593 enrolments, 7 course modules (AAA, BBB, CCC, DDD, EEE, FFF, GGG), 2013-2014

Licence: Creative Commons Attribution 4.0 International (CC BY 4.0)

Files:

FILE	ROWS	DESCRIPTION
studentInfo.csv	32,593	Student demographics and final results
studentAssessment.csv	173,912	Individual assessment submission scores
studentVle.csv	10,655,280	Daily VLE clickstream interactions
assessments.csv	206	Assessment metadata
studentRegistration.csv	32,593	Registration and withdrawal dates
vle.csv	6,364	VLE activity type metadata
courses.csv	22	Module and presentation metadata

Appendix D - System Inputs and Outputs

Input

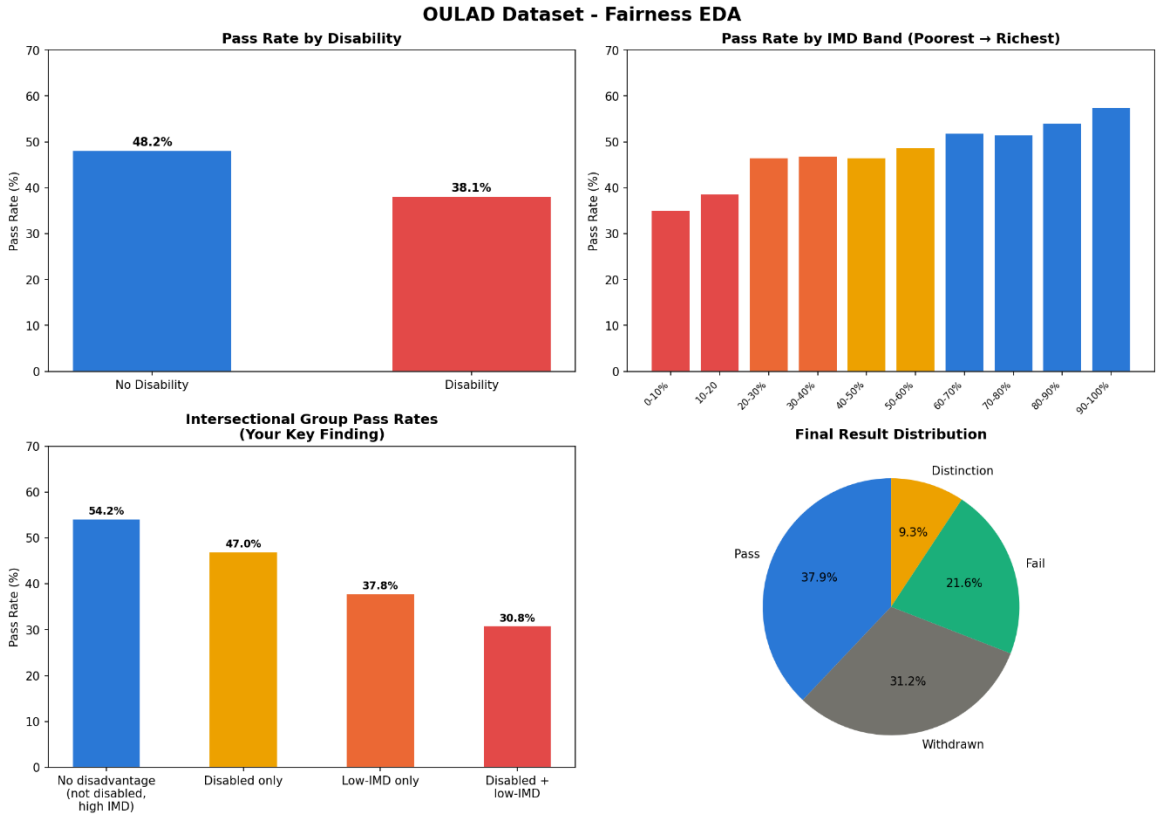
- archive.zip - compressed OULAD dataset (44 MB), uploaded to Google Colaboratory

Intermediate Output

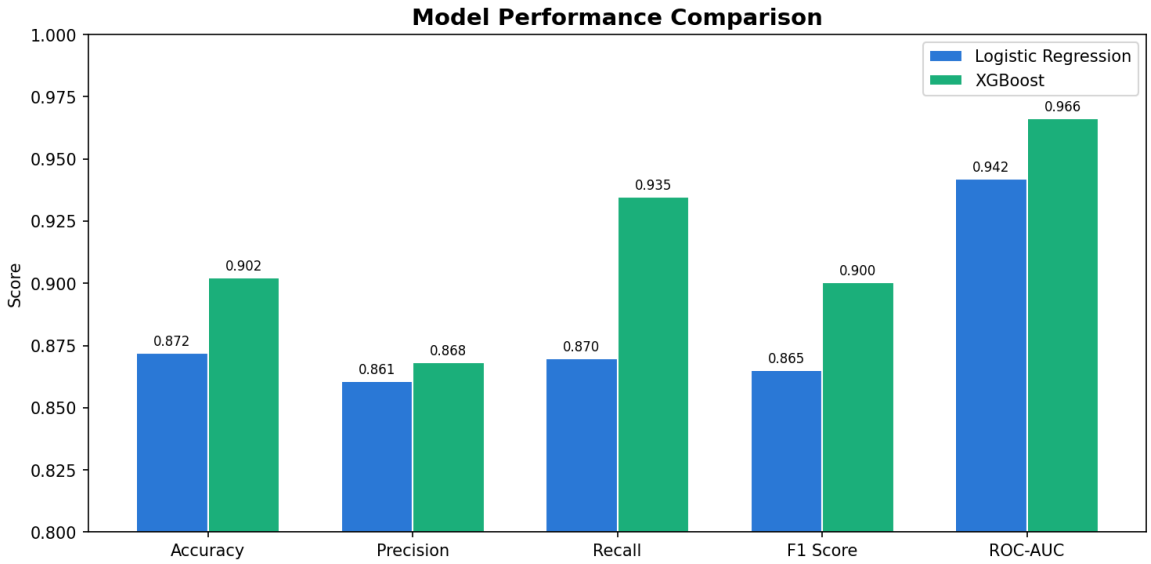
- oulad_clean.csv - merged and preprocessed dataset (32,593 rows x 22 columns, 3.6 MB)

Final Outputs

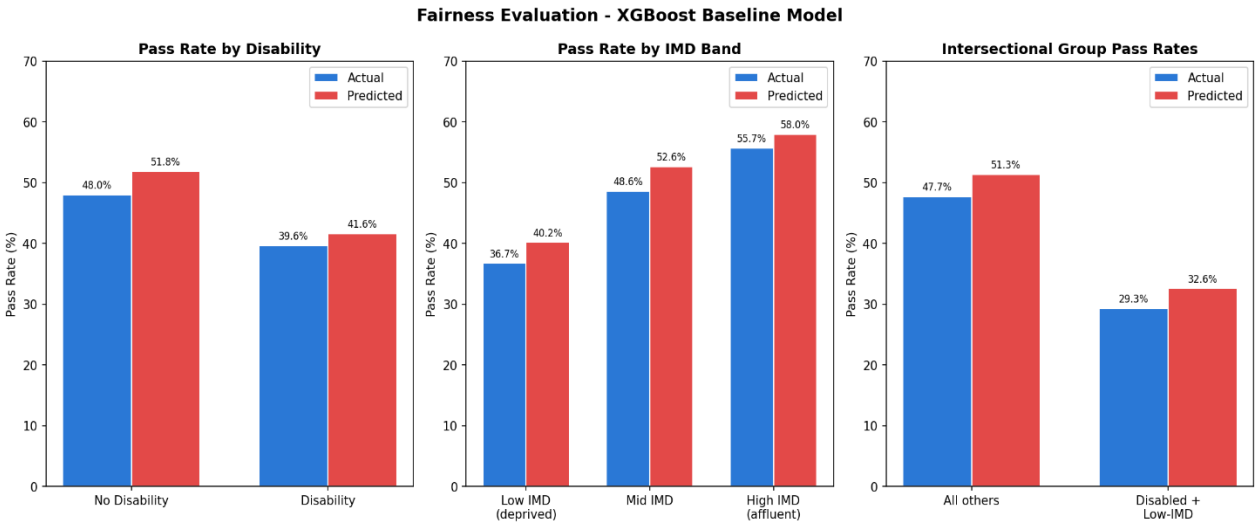
- ***eda_fairness_charts.png - exploratory fairness analysis (4 sub-charts: disability, IMD band, intersectional, outcome distribution)***



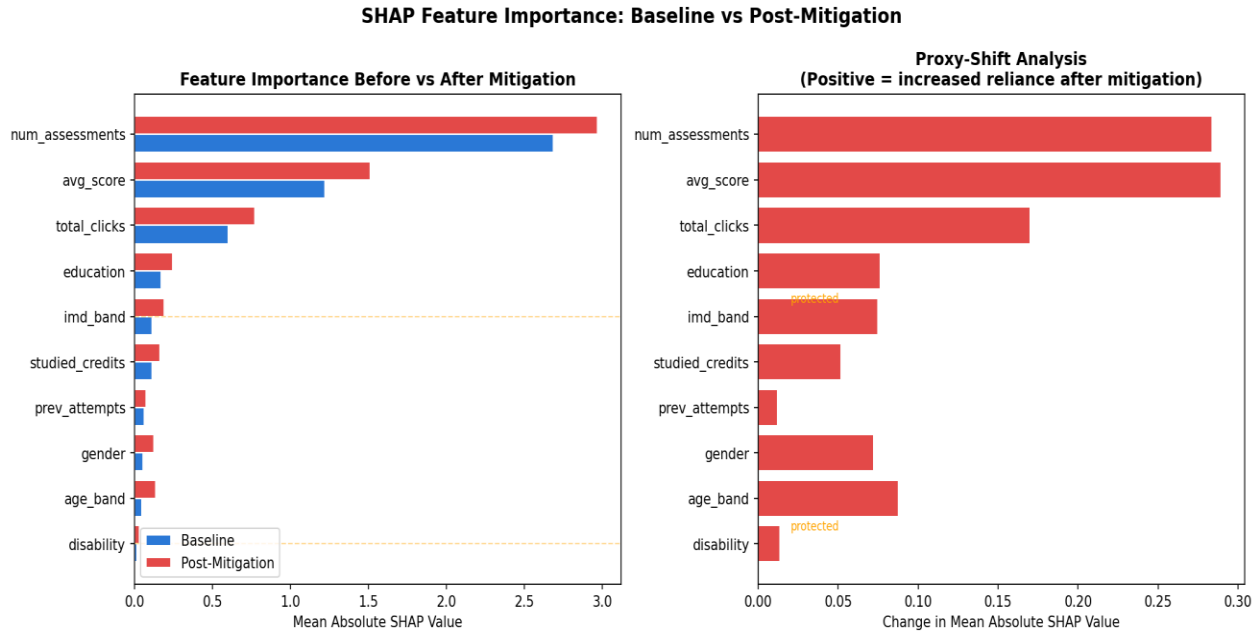
- ***model_comparison.png - Logistic Regression vs XGBoost comparison across five metrics***



- ***fairness_evaluation.png - actual vs predicted pass rates by three protected group configurations***



- ***shap_proxy_shift.png - SHAP feature importance baseline vs post-mitigation EG ensemble***



Appendix E - Key Results Summary

E.1 Model Performance

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	87.21%	86.07%	86.97%	86.52%	0.9421
XGBoost	90.23%	86.84%	93.47%	90.03%	0.9661

E.2 Baseline Fairness Metrics

Group	Predicted Pass Rate	DP Difference	EO Difference
Non-disabled	51.8%	0.102	0.044
Disabled	41.6%		
High IMD (affluent)	58.0%	0.178	0.028
Low IMD (deprived)	40.2%		
All others	51.3%	0.187	0.049
Disabled + Low-IMD	32.6%		

E.3 Post-Mitigation Results

Method	Sensitive Feature	EO Before	EO After	Improvement
Exponentiated Gradient	Disability	0.044	0.028	36.4%
Threshold Optimizer	Disability	0.044	0.007	84.1%
EG (IMD-trained)	IMD Band	0.028	0.018	35.7%
TO (IMD-trained)	IMD Band	0.028	0.025	10.7%

E.4 SHAP Proxy-Shift Summary

Feature	Baseline SHAP	Post-Mitigation SHAP	% Change
num_assessments	2.687	2.971	+10.6%
avg_score	1.222	1.512	+23.7%
total_clicks	0.602	0.772	+28.2%
imd_band (protected)	0.114	0.180	+57.9%
disability (protected)	0.017	1.429	+8,289%

All 10 features increased in SHAP value post-mitigation. The Exponentiated Gradient ensemble amplified rather than reduced protected attribute reliance - a previously undocumented fairness-washing failure mode.